

RKDF University

Program Outcomes, Program Specific Outcomes and Course Outcomes

Faculty of Science

Bachelor of Science

Programme Outcome (POs)

- PO1:** The B.Sc programme enabled the students to enhance their critical thinking, during the three year period of study and the curriculum stimulates the mental thoughts and assumptions of the students.
- PO2:** The B.Sc programme helps the students to take up practical work and compare the results with their assumptions, thereby leading to accuracy and validity of the practical knowledge.
- PO3:** The B.Sc programme leads to take decisions at intellectual, organizational and personal from different perspectives of life. Most of the students are getting selected for placements.
- PO4:** All the UG programmes have English and Hindi as compulsory Foundation course subjects. The motto behind is to improve the language skills like listening, speaking, reading and writing of all students in the class.
- PO5:** The B.Sc programme helps the students to develop better communication skills in English. The second language is generally the mother tongue i.e. Hindi. These subjects promoted culture, customs, moral and literary values in the students.

Program Specific Outcomes

- PSO1:** Bachelor of Science offers theoretical as well as practical knowledge about different subject areas.
- PSO2:** This course forms the basis of science for coherent understanding of the academic field to pursue multi and inter-disciplinary science careers in future. These subjects are Physics, Chemistry, Mathematics, Computer Science and Botany and Zoology.
- PSO3:** Able to plan and execute experiments or investigations, analyze and interpret data information collected using appropriate methods
- PSO4:** It helps to develop scientific temper and thus can prove to be more beneficial for the society as the scientific developments can make a nation or society to grow at a rapid pace through research.

PSO5: Think critically, follow innovations and developments in science and technology,

B.Sc. Mathematics Programme:

PSO1: Students acquired the Knowledge of Differential Equations of first order and first degree Orthogonal Trajectories, Differential Equations of the first order, Understand the Concept of the Plane, The Line, Sphere, Cones, Cylinders and Conicoids.

PSO2: Learned the concept of Groups, Subgroups,. Understand the Concept of Real Numbers, Sequences, Continuous Functions, Differentiation and Reimann Integration.

PSO3: Mathematical literacy in vector spaces, vector differentiation and vector integration.

PSO4: Learned to gain the Knowledge to apply the previous Knowledge on Integration & Differentiation to understand –Beta & Gamma functions and analyze the basic concepts of Bessel's equations, Hermite Polynomials, Laguerre Polynomials and Legendre's equation.

BMM-111 Algebra and Trigonometry

Course outcome:

- CO1: Understand the concepts of vector spaces, subspaces, bases, dimension and their properties.
- CO2: Use computational techniques and algebraic skills essential for the study of systems of Linear equations, matrix algebra, vector spaces, eigenvalues and eigenvectors, Orthogonality and Diagonalization. (Computational and Algebraic Skills).
- CO3: Discuss the behaviour of the Trigonometrical function.
- CO4: Find Boolean Algebra and graph theory..
- CO5: Expand $\sin n\theta$, $\cos n\theta$ and $\tan n\theta$ by using Demoivre's theorem.
- CO6: Expand $\cos n\theta$, $\sin n\theta$ and $\tan n\theta$ in terms of θ .

BMM-121 Calculus and Numerical Analysis

Course outcome:

- CO1: Assimilate the notions of limit of a sequence and convergence of a series of real numbers.
- CO2: Calculate the limit and examine the continuity of a function at a point.
- CO3: Understand the consequences of various mean value theorems for differentiable functions equations.
- CO4: Sketch curves in Linear Differential equations and equations reducible to the linear form.
- CO5: Apply derivative tests in optimization problems in numerical integration.
- CO6: Student will be to understand differentiation and fundamental theorem in differentiation and various rules.

BMM-131 Linear Algebra

Course outcome:

- CO1: Understand the concepts of vector spaces, subspaces, bases, dimension and their properties.
- CO2: Relate matrices and linear transformations; compute eigen values and eigen vectors of linear transformations.
- CO3: Learn properties of inner product spaces and determine orthogonality in inner product spaces.
- CO4: Realise importance of adjoint of a linear transformation and its canonical form.

BMM-141 Analysis

Course outcome:

- CO1: To apply appropriate numerical methods to solve the problem with most accuracy.
- CO2: Using appropriate numerical methods determine approximate solution of ODE and system of linear equation.
- CO3: Compare different methods in numerical analysis w.r.t accuracy and efficiency of solution.
- CO4: Find numerical solutions of system of Cauchy's theorem and check the accuracy of the solutions.

BMM-151 Real Analysis – I

Course Outcome:

- CO1: Find numerical solutions of system of linear equations and check the accuracy of the solutions.
- CO2: Solve initial and boundary value problems in differential equations using numerical methods.
- CO3: Apply various numerical methods in real life problems.
- CO4: Describe fundamental properties of the real numbers that lead to the formal development of real analysis.
- CO5: Demonstrate an understanding of limits and how they are used in sequences, series, Construct rigorous mathematical proofs of basic results in real analysis.

BMM-161 Programming Language with “C” Theory

Course outcome:

- CO1: Understand and apply the programming concepts of C++ which is important for mathematical investigation and problem solving.
- CO2: Use mathematical libraries for computational objectives.
- CO3: Represent the outputs of programs visually in terms of well formatted text and plots.

B.Sc. Physics Programme:

- PSO1:** Apply the basic laws of physics in the areas of classical mechanics, Newtonian gravitation, Special Relativity, electromagnetism, geometrical and physical optics, thermodynamics and statistical mechanics.
- PSO2:** Analyze the quantum methods in the solution of problems involving atomic spectra, blackbody radiation, the photoelectric effect, X-ray emission, the structure of the atom, and one-dimensional potentials.
- PSO3:** Understand classical experimental techniques and modern measurement technology including analog and digital electronics, laboratory test equipment, optics, lasers, and detectors.
- PSO4:** The student develop good experimental technique, including proper setup and care of equipment, conducting experiments and analyzing results and make meaningful comparisons between experiment and theory.

BPY 111 MECHANICS

Course outcome:

- CO1: Relative motion. Inertial and non inertial reference frames.
- CO2: Parameters defining the motion of mechanical systems and their degrees of freedom.
- CO3: Study of the interaction of forces between solids in mechanical systems.
- CO4: Centre of mass and inertia tensor of mechanical systems.
- CO5: Application of the vector theorems of mechanics and interpretation of their results.
- CO6: Newton's laws of motion and conservation principles.
- CO7: Introduction to analytical mechanics as a systematic tool for problem solving. 8 Use of mechanical simulation software.

BPY 121 PROPERTIES OF MATTER AND SOUND**Course outcome:**

- CO1: Study the elastic behavior of materials
- CO2: Analyze the relationship between various types of experiments
- CO3: Perform the procedure as per standard values
- CO4: Understand the applications

BPY 131 ELECTRICITY AND MAGNETISM**Course outcome:**

- CO1: Study the electric field using coulombs inverse square law in electrostatics of current
- CO2: Analyse the chemical and heating effect of current
- CO3: Analyse the relations between b , h and m
- CO4: Understand the faradays laws of electromagnetic induction by rayleigh's method
- CO5: Analyse the value of maxwell equation- boundary conditions

BPY 141 ELECTRICITY AND MAGNETISMBASIC ELECTRONICS**Course outcome:**

- CO1: Apply knowledge of electricity and magnetism to explain natural physical processes and related technological advances.
- CO2: Use an understanding of calculus along with physical principles to effectively solve problems encountered in everyday life, further study in science.
- CO3: Predict the behavior of any electrical and magnetic circuits.
- CO4: Formulate and solve complex AC, Dc circuits.
- CO5: Identify the type of electrical machine used for that particular application

BPY 151 NUCLEAR PHYSICS**Course outcome:**

- CO1: Understand the properties of positive rays, experimental proof by frank and hertz method
- CO2: Analyse the relationship between various types of couplings
- CO3: Understand the properties of x-ray s verification
- CO4: Analyse the ideas of basics of nucleus and their energy
- CO5: Perform the procedures for nuclear fission and fusion

BPY 161 ELECTRONIC COMMUNICATIONS**Course outcome:**

- CO1: Able to design and implement various digital modulation and demodulation techniques.
- CO2: Able to identify and describe different techniques in modern digital

- communications, in particular in source coding using MATLAB or similar tools.
- CO3:** Able to understand and verify sampling theorem for practical applications.
- CO4:** Able to implement and verify DFT property using MATLAB
- CO5:** Able to comment on Stability and Causality of Discrete time system using Z-transform on MATLAB tool.
- CO6:** Able to design and implement digital filter and use of various windows

B.Sc. Chemistry Programme:

- PSO1:** Understand the structural difference among solids, liquids, gases and solutions
- PSO2:** Gained the Knowledge in electrical and thermal properties of d- block elements through Free Electron Theory, Valency Bond Theory and Band Theory
- PSO3:** Analyze the structures of glucose, fructose and sucrose and their chemical properties
- PSO4:** Use spectrophotometer to analyze known and unknown organic and natural compounds.

BCH-111 BASED ON INORGANIC, ORGANIC AND PHYSICAL CHEMISTRY

Course outcome:

- CO1: To learn about the covalent bond and its applications.
- CO2: Learn about the various critical phenomenon and its application.
- CO3: Derivation of Vander Waal's Equation of State and gaseous states.
- CO4: To learn about the Concept of isomerism and its brief idea.
- CO5: To study about the Mechanism of Organic Reactions and its details.

BCH-121 Based on inorganic, organic and physical chemistry

Course outcome:

- CO1: To learn about the Rate of reaction, rate equation and its explanations of its reactions.
- CO2: Explanations of metallic bond and semiconductors.
- CO3: To study about the various theory and about alkenes.
- CO4: To learn about the different laws and its applications.

BCH-131 Based on inorganic, and Physical Chemistry

Course outcome:

- CO1: To study about the elements of IInd and IIIrd transition series and its details.
- CO2: To learn about the thermodynamics and its laws and applications.
- CO3: To know about the carbolic acid and alcohols and epoxides.

BCH-141 BASED ON INORGANIC, AND PHYSICAL CHEMISTRY

Course outcome:

- CO1: To learn about the details of lanthanide compounds and actinides.
- CO2: To describe Electrolytic and Galvanic cells.
- CO3: Students will be able to explain the quantitative relationship between T, V, n & P as described by kinetic molecular theory.
- CO4: To study about the carbonyl group.

BCH-151 BASED ON INORGANIC, ORGANIC & PHYSICAL CHEMISTRY

Course outcome:

- CO1: To study about the monosaccharide and its details.
- CO2: To describe about the transition metal complex of the various properties.

- CO3: Various spectroscopy methods and spectrums.
- CO4: To learn about organo compounds.

BCH-161 Based on inorganic, Organic & Physical Chemistry

Course outcome:

- CO1: To learn about the bioinorganic chemistry and its details.
- CO2: To different laws and equations of radiation.
- CO3: To study about the hetrocyclic compounds.
- CO4: To learn about the various biomolecules , classification and in details.

B.Sc. Botany Programme:

PSO I: The students gained the knowledge relating to the interrelations hips of different plant groups and their evolutionary tendencies

PSO2: The students acquired the knowledge enough to differentiate animals plants and Microorganisms with their distinguishing characteristics.

PSO3: Student of the gained the Knowledge and acquired capability of applying the Knowledge in the areas of like Agriculture, Plant Medicines, Horticulture and Tissue culture.

BSB 111 Diversity of Microbes & Cryptogams

Course Outcome:

- CO1: Understand the diversity among Bacteria, Viruses and Algae. Know the systematic, morphology and structure, of Bacteria, Viruses and Algae.
- CO2: Understand the life cycle pattern of Bacteria, Viruses and Algae.
- CO3. Understand the useful and harmful activities of Bacteria, Viruses and Algae
- CO4. Understand the morphological diversity of Bryophytes and Pteridophytes.
- CO5. Understand the economic importance of the Bryophytes and Pteridophytes.
- CO6. Know the evolution of Bryophytes and Pteridophytes

BSB 121 Cell Biology & Genetics

Course Outcome:

- CO1. Students will understand the structures and purposes of basic components of prokaryotic and eukaryotic cells, especially macromolecules, membranes, and organelles
- CO2. Students will understand how these cellular components are used to generate and utilize energy in cells
- CO3. Students will understand the cellular components underlying mitotic cell division.
- CO4. Students will apply their knowledge of cell biology to selected examples of changes or losses in cell function. These can include responses to environmental or physiological changes, or alterations of cell function brought about by mutation.

BSB 131 Diversity & systematics of seed plants

Course Outcome:

- CO1: Know the conceptual development of taxonomy & systematics. Understand the general range of variations in the group of angiosperms.
- CO2: Trace the history of development of systems of classification emphasizing angiospermic taxa.
- CO3: Learn about the characters of biologically important families of angiosperms.
- CO4: Know the floral variations in angiosperm families, their phylogeny and evolution.

- Understand various rules, principles and recommendations of plant nomenclature.
- CO5: Know modern trends in taxonomy.
- CO6: Understand major evolutionary trends in various parts of angiosperm plants. Students will understand the structures and purposes of basic components of prokaryotic and eukaryotic cells, especially macromolecules, membranes, and organelles

BSB 141 Structure, development & reproduction in flowering plants

Course Outcome:

- CO1: Study of the families with respect to morphological characters using botanical terms, floral formula, floral diagram and classification giving.
- CO2: Identification of genus and species with the help of flora of the plant materials.
- CO3: Preparation of artificial, bracketed/indented dichotomous keys based on vegetative and reproductive characters.
- CO4: Study of Morphological and biological peculiarities of plants.

BSB 151 Plant physiology and biochemistry

Course Outcome:

- CO1: Know importance and scope of plant physiology.
- CO2: To understand the plants and plant cells in relation to water.
- CO3: Understand the process of photosynthesis in higher plants with particular emphasis on light and dark reactions, C3 and C4 pathways.
- CO4: Understand the respiration in higher plants with particular emphasis on aerobic and anaerobic respiration.
- CO5: Understand the biochemical nature of nucleic acids, their role in living systems, experimental evidences to prove DNA as a genetic material.
- CO6: Understand the process of synthesis of proteins and role of genetic code in polypeptide formation

BSB 161 Plant ecology, biodiversity and phytogeography

Course Outcome:

- CO1: Understand the “Science of Heredity”. Realize the role of genes in evolution of species.
- CO2: To understand linkage, segregation and mutation of genes during evolution.
- CO3: Understand the science of plant breeding.
- CO4: To introduce the student with branch of plant breeding for the survival of human being from starvation. To study the techniques of production of new superior crop varieties.
- CO5: To study the evolution in living organisms.

B.Sc. Zoology Programme:

- PSO1 :** Learned about Ecological adaptations and the knowledge of inter relationships of the animal with other groups like plants and microbes.
- PSO2 :** The students gained practical knowledge about the specimen observation and its classification depending on the observed characteristics, Determination of PH , Alkalinity, concentration of oxygen in the given water sample and also analysis of human blood regarding types of blood groups(A,B,AB,O), Estimation of Haemoglobin, total count of RBC ,WBC etc.
- PSO3:** The students understood the applications of Zoology in Aquaculture, Vermiculture, Sericulture, Poultry Science and Fundamentals of Clinical Science and Immunology.

BSZ 111 INVERTEBRATA & CHORDATA

Course outcome:

- CO1: To classify Phylum Porifera with taxonomic Keys.
- CO2: To describe the Phylum Coelenterata and its Polymorphism.
- CO3: To explain classification of protozoa and diseases caused by them.
- CO4: To explain general characters of Amphioxus.
- CO5: To identify the characters of Amphibia and its parental care.

BSZ 121 CELL BIOLOGY & GENETICS

Course outcome:

- CO1: Describe cell cycles and its regulation.
- CO2: Write down molecular biology techniques.
- CO3: Explain causes and role of extinction in evolution.
- CO4: To identify chromosomal mutations and in borne errors of metabolism.
- CO5: To describe differences between prokaryotic and Eukaryotic cells.

BSZ 131 Animal Physiology & Developmental Biology & Immunology

Course outcome:

- CO1: Identify various stages of embryological development of Amphioxus, frog.
- CO2: To study the difference types of nutrition.
- CO3: To learn about the excretion of organisms.
- CO4: To study about spermatogenesis.

BSZ 141 ECOLOGY & EVOLUTIONARY BIOLOGY

Course outcome:

- CO1: To Describe Environmental Population and its control measures.
- CO2: To understand methods of wildlife and conservation and endangered species.
- CO3: To describe Innate and Acquired types of behavior.
- CO4: Knowledge regarding principles, applications and management of environmental science.
- CO5: To study of the ecosystem of pond.

BSZ 151 MICROBIOLOGY AND BIOCHEMISTRY

Course outcome:

- CO1: To study the classification of different microorganisms.
- CO2: To learn about the different morphology of water, soil, food microbiology and its effects.
- CO3: Learn the structure and functions of bio-molecules and their role in metabolism.

BSZ 161 Biostatistics & Computer Applications & Basic Animal Biotechnology

Course outcome:

- CO1: To study of the data types and it's tabular and graphical representation.
- CO2: To learn about the central tendency and its different methods and equations.
- CO3: To learn about the computer and its application to biology.
- CO4: To learn the Cryo preservation of cell cultures and animal cell culture.

B.Sc. Biotechnology Programme:

PSO1: Identify DNA structure and replication, transcription, translation and gene expression.

PSO2 : Learned about Ecological adaptations and the knowledge of inter relationships of the animal with other groups like plants and microbes.

PSO3 : The students gained practical knowledge about the specimen observation and its classification depending on the observed characteristics, Determination of PH , Alkalinity, concentration of oxygen in the given water sample and also analysis of human blood regarding types of blood groups(A,B,AB,O) etc.

PSO4: The students understood the applications of Biotechnology in Aquaculture, Vermiculture, Sericulture, Poultry Science and Fundamentals of Clinical Science and Immunology.

BBT-111 CELL AND MOLECULAR BIOLOGY & MOLECULAR GENETICS

Course outcome:

- CO1: Exhibit a knowledge base in genetics, cell and molecular biology, and anatomy and physiology
- CO2: An introduction to the physical and chemical organization of living organisms; cell structure, function, and metabolism;
- CO3: Classical and molecular genetics; gene regulation; genetic engineering;
- CO4: Molecular aspects of development; and reproduction.

BBT-121 BIOMOLECULES & ORGANIC MECHANISMS IN BIOLOGY

Course outcome:

- CO1: Through this course the students are exposed to importance of biological macromolecules.
- CO2: They acquire knowledge in the quantitative and qualitative estimation of biomolecules.
- CO3: They study the influence and role of structure in reactivity of biomolecules.
- CO4: At the end of the course, the students have a thorough understanding on the role of biomolecules and their functions.

BBT-131 FOOD BIOTECHNOLOGY & BIOPROCESS ENGINEERING

Course outcome:

- CO1: This course enables the students to understand application of biotechnology in Food processing industries.
- CO2: To List out the commercially important enzyme used in food industries.
- CO3: To make the student to understand the causes of food spoilage, control and preventive measures for harmful microorganism.
- CO4: At the end of this course, the student will have an idea of food constituents, importance and their daily dietary allowances; scope and prospects for food industries.

BBT-141 MOLECULAR IMMUNOLOGY & RECOMBINANT DNA TECHNOLOGY

Course outcome:

- CO1: Understanding the concept of Gene and the gene architecture.
- CO2: Overview of the central dogma of life and various molecular events.
- CO3: Understanding the principles and applications of Polymerase Chain Reaction(PCR).
- CO4: Molecular Events of Transcription and processing of transcripts, RNA editing.

CO5: Learn on assigning gene function through mutagenesis and genetic engineering.

BBT-151 INSTRUMENTATION & BASIC BIOSTATISTICS

Course outcome:

- CO1: Understanding the principles and operation of ELISA readers; Spectrophotometers; Freeze driers/lyophilizes.
- CO2: Learning principle and operations of Thermal cyclers; Real-time PCR; DNA synthesizer; DNA sequencer; Microscopes: Light, stereo, phase contrast and inverted.
- CO3: Learning various analytical techniques and principles for separation of Biopharmaceuticals including Ion exchange chromatography, Affinity chromatography, Gel filtration chromatography, HPLC method for purification of proteins.
- CO4: To learn the Methodologies of Biostatistics and its application in selection of the Biological samples.
- CO5: To understand Biostatistical Analysis of the Biological Experiments.

BBT-161 BIOINFORMATICS & TISSUE CULTURE TECHNOLOGY

Course outcome:

- CO1: To have knowledge on Biomolecules, their importance and Classification I Bioinformatic.
- CO2: To explain the properties of Biomolecules.
- CO3: To understand principles of animal culture, media preparation.
- CO4: To explain Invitro fertilization and embryo transfer technology.
- CO5: To describe meristem culture and clonal propagation of plants on a commercial scale.

B.Sc. Microbiology Programme:

PSO I: The students gained the knowledge relating to the interrelationships of different microorganisms groups and their evolutionary tendencies

PSO2: The students acquired the knowledge enough to differentiate animal's plants and microorganisms with their distinguishing characteristics.

PSO3: Student of the gained the Knowledge and acquired capability of applying the Knowledge in the areas of like Agriculture, Plant Medicines, Horticulture and Tissue culture.

PSO4: Comprehend the various methods for identification of unknown microorganisms

BMB 111 Fundamental of Microbiology

Course outcome:

- CO1: Demonstrate theory and practical skills in microscopy and their handling techniques and staining procedures
- CO2: Understand the basic microbial structure and function and study the comparative characteristics of prokaryotes and eukaryotes and also
- CO3: Understand the structural similarities and differences among various physiological groups of bacteria/archaea
- CO4: Know various Culture media and their applications and also understand various physical and chemical means of sterilization
- CO5: Know General bacteriology and microbial techniques for isolation of pure cultures of bacteria, fungi and algae
- CO6: Master aseptic techniques and be able to perform routine culture handling tasks safely and effectively

BMB 121 Microbial Physiology & Microbial Genetics

Course outcome:

- CO1: Know the terms and terminologies related to molecular biology and microbial growth.
- CO2: Understand the properties, structure and function of genes in living organisms at the molecular level
- CO3: Explain the significance of central dogma of gene action
- CO4: Have a conceptual knowledge about DNA as a genetic material, enzymology, and replication strategies
- CO5: Understand the molecular mechanisms involved in transcription and translation

BMB 131 Microbial Biochemistry & Environmental Microbiology

Course outcome:

- CO1: Describe the concepts of electrolytes and electrolytic dissociation, pH and its biological significance, buffers, Henderson-Hasselbalch equation, biological buffer systems and their importance.
- CO2: Understanding the laws of thermodynamics , concepts of entropy, enthalpy and free energy changes and their application to biological systems and various biochemical studies and reactions.
- CO3: Conceptual knowledge of air, food, water and soil microbiology.
- CO4: Overview of major biomolecules –carbohydrates, lipids, proteins, aminoacids, nucleic acids, classification, structure, function of the above mentioned biomolecules
- CO5: Discuss the biosynthesis and the degradation pathways involved.
- CO6: Specify the biological significance of biomolecules in metabolism
- CO7: Conceptual knowledge of properties, structure, function of enzymes, enzyme kinetics and their regulation ,enzyme engineering, Application of enzymes in large scale industrial processes

BMB 141 IMMUNOLOGY & MEDICAL MICROBIOLOGY

Course outcome:

- CO1: Demonstrate an understanding of key concepts in immunology
- CO2: Understand the overall organization of the immune system
- CO3: Conceptualize how the collection of individual clones of lymphocytes (termed the “immune repertoire”) arises from rearrangement within two genetic loci: the Ig gene in B cells and the antigen receptor in T cells.
- CO4: Learn how “clonal selection” allows for the expansion of a limited number of antigen-recognizing lymphocytes in response to an specific antigenic stimulus
- CO5: Begin to appreciate the significance of maintaining a state of immune tolerance sufficient to prevent the emergence of autoimmunity
- CO6: To make them understand the salient features of antigen antibody reaction & its uses in diagnostics and various other studies.

BMB 151 BIOINFORMATICS, BIOSTATISTICS & INDUSTRIAL MICROBIOLOGY

Course outcome:

- CO1: To study the bioinformatics and its application on biology.
- CO2: To know the statistical application in biology.
- CO3: To understand the industrial microbiology and its process.
- CO4: To study the scale up process.

BMB 161 ANALYTICAL MICROBIOLOGY & APPLIED MICROBIOLOGY

Course outcome:

- CO1: To understand the bioassay and quality control.
- CO2: To study the classification of different microorganisms.
- CO3: To learn about the different morphology of agriculture microbiology and its effects.
- CO4: To know the different types of colorimetry , spectrophotometry and separation techniques.

B.Sc. Food Technology Programme:

PSO I: The students gained the knowledge relating to the food and different microorganisms groups and their evolutionary tendencies with relation to the food technologies.

PSO2: The students acquired the knowledge enough to differentiate animal's plants and microorganisms with their distinguishing characteristics.

PSO3: Student of the gained the Knowledge and acquired capability of applying the Knowledge in the areas of like Agriculture, Plant Medicines, Horticulture and Tissue culture.

PSO4: Comprehend the various methods for identification of unknown microorganisms

BFS 111 INTRODUCTION TO FOOD TECHNOLOGY

Course outcome:

- CO1: To understand the relationship between food, nutrition and health.
- CO2: To understand the history and the concept of food science.
- CO3: To know the classification of food.
- CO4: To study about the food adulteration and food additive.
- CO5: To know the important nutrients of body in details.

BFS 121 BASIC NUTRITION

Course outcome:

- CO1: To understand the relationship between food, nutrition and health.
- CO2: To understand digestion, absorption, functions and food sources of various nutrients
- CO3: To appreciate the concept of balanced and healthy diets.
- CO4: To know the different methods of cooking and ways to prevent nutrient losses.
- CO5: To be able to plan and prepare meals and nutritious dishes for various age groups.
- CO6: To be able to assess nutritional status of adults

BFS 131-BASIC PRINCIPLES OF FOOD ENGINEERING

Course outcome:

- CO1: Understand cold preservation ,Freezer types and functioning
- CO2: Understand Dehydration, Dryer types and functioning
- CO3: Understand the engineering units
- CO4: To know the process of heat transfer in food.

BFS 141- BIOCHEMISTRY

Course outcome:

- CO1: Understand and describe the general chemical structures of the major components of foods (water, proteins, carbohydrates, and lipids).
- CO2: Give a molecular rationalization for the observed physical properties and reactivity of major food components.

- CO3: Provide a theoretical explanation for observed extent and rates of reactions that are common to foods. Predict how processing conditions are likely to change the reactivity of food components.
- CO4: To predict how changes in overall composition are likely to change the reactivity of individual food.
- CO5: To know the different metabolisms of nutrients.

BFS 151-FOOD MICROBIOLOGY

Course outcome:

- CO1: Understand the important genera of microorganisms associated with food and their characteristics, their growth pattern and parameters.
- CO2: Comprehend the role of the microorganisms in spoilage of foods and methods of their control.
- CO3: Knowledge about the beneficial role of microorganisms and different types of fermented foods.
- CO4: Identify the role of microorganisms in food borne diseases and control measures.
- CO5: Understand the laboratory techniques to detect, quantify, and identify microorganisms in foods.

BFS 161 - BIOPROCESS ENGINEERING AND TECHNOLOGY

Course outcome:

- CO1: To know the about the industrial microorganism.
- CO2: To understand the types of fermentation processes.
- CO3: To get a brief knowledge about food technology.

B.Sc. Genetics Programme:

- PSO1:** Identify DNA structure and replication, transcription, translation and gene expression.
- PSO2 :** Learned about Ecological adaptations and the knowledge of inter relationships of the animal with other groups like plants and microbes.
- PSO3 :** Learn on assigning gene function through mutagenesis and genetic engineering.
- PSO4:** The students understood the applications of Biotechnology in Aquaculture, Vermiculture, Sericulture, Poultry Science and Fundamentals of Clinical Science and Immunology.

BGN-111 FUNDAMENTALS OF CELL BIOLOGY

Course outcome:

- CO1: Exhibit a knowledge base in genetics, cell and molecular biology, and anatomy and physiology.
- CO2: An introduction to the physical and chemical organization of living organisms; cell structure, function, and metabolism;
- CO3: Classical and molecular genetics; gene regulation; genetic engineering;
- CO4: Molecular aspects of development; and reproduction.
- CO5: Understanding of cell cycle.

BGN-121 PRINCIPLES OF GENETICS

Course outcome:

- CO1:** Students will understand the structures and purposes of basic components of

prokaryotic and eukaryotic cells, especially macromolecules, membranes, and organelles

- CO2:** Students will understand how these cellular components are used to generate and utilize energy in cells
- CO3:** Students will understand the cellular components underlying mitotic cell division.
- CO4:** Students will apply their knowledge of cell biology to selected examples of changes or losses in cell function. These can include responses to environmental or physiological changes, or alterations of cell function brought about by mutation.

BGN-131 CYTOGENETICS

Course outcome:

- CO1:** Understanding the concept of Gene and the gene architecture.
- CO2:** Overview of the central dogma of life and various molecular events.
- CO3:** Understanding the principles and applications of Polymerase Chain Reaction(PCR).
- CO4:** Molecular Events of Transcription and processing of transcripts, RNA editing.
- CO5:** Learn on assigning gene function through mutagenesis and genetic engineering.
- CO6:** Drosophila genetics its importance as model for human genetic disorders,
- CO7:** Mutagenesis & isolation of new variants

BGN-141 MOLECULAR GENETICS

Course outcome:

- CO1:** Molecular aspects of development; and reproduction.
- CO2:** Identify DNA structure and replication, transcription, translation and gene expression.
- CO3:** Molecular Events of Transcription and processing of transcripts, RNA editing.
- CO4:** Learn on assigning gene function through mutagenesis and genetic engineering.

BGN-151 RECOMBINANT DNA TECHNOLOGY

Course outcome:

- CO1:** Understanding the concept of Gene and the gene architecture.
- CO2:** Overview of the central dogma of life and various molecular events.
- CO3:** Understanding the principles and applications of Polymerase Chain Reaction(PCR).
- CO4:** Molecular Events of Transcription and processing of transcripts, RNA editing.

BGN-161 APPLIED AND BEHAVIORAL GENETICS

Course outcome:

- CO1:** The significance of genetic science to human society.
- CO2:** The fundamental principles and functions of Genetic science.
- CO3:** The divisions in a DNA fingerprinting laboratory.
- CO4:** The working of the bioinformatic.
- CO5:** The importance of GeneticresourcesandBiodiversity.
- CO6:** The causes of behavior.

B.Sc. Computer Science Programme:

PSO1: To learn about the basic of computer , about its history and its invention in details.

PSO2 : To learn about the different function like hash and knowledge about IP address.

PSO3 : Learn on assigning gene function through mutagenesis and genetic engineering with the help of computers

PSO4: Develop architectural design, and implement by following coding principles.

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BCS-111 COMPUTER FUNDAMENTALS & APPLICATION

Course outcome:

CO1: To study about the basic of computer , about its history and its invention in details.

CO2: To learn about the hardware and the software system of the computer.

CO3: To know about the network , its works and internet in details.

BCS-121 Operating System

Course outcome:

CO1: To study about the system porgraming and its opareting system, its history.

CO2: To learn the files, process and others .

CO3: Explanation of memory managements in details.

CO4: Design a model internet with various categories of networks and test the transmission rate

BCS-131 NETWORK SECURITY

Course outcome:

CO1: To study the security trends and cryptography.

CO2: To know about the Trojans and back doors in details.

CO3: To study about the different function like hash and all.

CO4: Knowledge about IP in details.

BCS-141 SOFTWARE DESIGNING

Course details:

CO1: Learn the phases of software development.

CO2: Develop process models and process system models.

CO3: Gather, understand, analyze and specify requirements

BCS-151 COMPUTER NETWORK

Course outcome:

CO1: Create a new protocol and test its efficiency.

CO2: Design a new network architecture using protocols and interfaces.

CO3: Create a hybrid topologies using the existing topologies, and check in efficiency.

CO4: Apply different encoding and decoding mechanisms involved in different types of transmission media and to measure the transmission impairments.

BCS-161 INFORMATION STORAGE AND MANAGEMENT

Course outcome:

- CO1: To learn about the storage technology and networked storages.
- CO2: To know about the storage systems and architecture.
- CO3: To learn about the hybrid storage solutions.

B.Sc. Electronics Programme:

- PSO1:** To give knowledge to the students of working of some power electronic devices, transducers and application of transducers..
- PSO2 :** Analyze the data transfer information through serial & parallel ports.
- PSO3 :** Make students understand the transmission of voice and data through various networks.
- PSO4:** Learn to Design & implement a triggering / gate drive circuit for a power device like SCR, MOSFET etc.
- PSO5:** To study the analysis and synthesis of TV Pictures, Composite Video Signal, Receiver, Picture Tubes and Television Camera Tubes.

BEC 111 FUNDAMENTALS OF ELECTRONICS

Course outcome:

- CO1: To give knowledge of some basic electronic components and circuits.
- CO2: To introduce basics of diode and transistor circuits.
- CO3: To understand working of some I C based circuits.
- CO4: To introduce basic aspect of electronic communication systems.

BEC 121 SEMICONDUCTOR DEVICES AND IC FABRICATION TECHNOLOGY

Course outcome:

- CO1: Understand the characteristics of IC and Op-Amp and identify the internal structure.
- CO2: Derive and determine various performances based parameters and their significance for Op-Amp.
- CO3: Use the basic logic gates and various reduction techniques of digital logic circuit in detail.
- CO4: Understand the basic concept of probability, random variables & random signals and develop the ability to find correlation, CDF, PDF and probability of a given event.
- CO5: Understand and apply the functionalities of PLL to Frequency synthesizer, multiplier, FM, and AM demodulators.

BEC 131 ELECTRONIC COMMUNICATION SYSTEMS

Course outcome:

- CO1: Explain and apply the concepts telecommunication switching, traffic and networks.
- CO3: Explain and apply concepts of GSM and CDMA system.
- CO3: Carry out Link power budget and Rise Time Budget by proper selection of components and check its viability.
- CO4: Carry out Satellite Link design for Up Link and Down Link.

BEC 141 MICROPROCESSOR AND ITS APPLICATIONS

Course outcome:

- CO1: Recall and apply a basic concept of digital fundamentals to Microprocessor based

personal computer system.

CO2: Identify a detailed s/w & h/w structure of the Microprocessor.

CO3: Illustrate how the different peripherals (8255, 8253 etc.) are interfaced with Microprocessor.

CO4: Distinguish and analyze the properties of Microprocessors & Microcontrollers.

BEC 151 COMMUNICATIONS

Course outcome:

CO1: To study audio recording systems such CD/DVD recording, Audio Standards, and Acoustics principles.

CO2: To study the various Colour Television systems with a greater emphasis on television standards.

CO3: Able to know standing wave ratio for open, short and matched terminations on trainer kit.

CO4: Explain and apply the concepts telecommunication switching, traffic and networks.

CO5: Carry out Link power budget and Rise Time Budget by proper selection of components and check its viability.

BEC 161 Electronic Instrumentation, Digital System Design & Microcontroller

Course outcome:

CO1: Learn importance of microcontroller in designing embedded application.

CO2: Learn use of hardware and software tools.

CO3: Develop interfacing to real world devices.

CO4: Able to design and implement various digital modulation and demodulation techniques.

CO5: Design and implementation of Basic Microcontroller Based system using 8051 and PIC Microcontroller.

B.Sc. Forensic Science Programme:

PSO1: The learn significance of forensic science to human society and the fundamental principles and functions of forensic science..

PSO2: The give students knowledge about critical assessment of advanced forensic techniques like polygraph, narco analysis and brain electrical oscillation signatures

PSO3: Learn on assigning gene function through mutagenesis and genetic engineering

PSO4: The learn about consequences of crime in society and the elements of criminal justice system.

BFC-111 Introduction to Forensic Science, Crime and Society

Course outcome:

CO1: The divisions in a forensic science laboratory.

CO2: The working of the forensic establishments in India and abroad.

CO3: The importance of criminology.

- CO4: The causes of criminal behavior.
- CO5: The significance of criminal profiling to mitigate crime.

BFC-121 Criminal Law and Forensic Psychology

Course outcome:

- CO1: Elements of Criminal Procedure Code related to forensic science.
- CO2: Acts and provisions of the Constitution of India related to forensic science.
- CO3: Acts governing socio-economic crimes.
- CO4: Acts governing environmental crimes.
- CO5: The overview of forensic psychology and its applications.
- CO6: The legal aspects of forensic psychology.
- CO7: The significance of criminal profiling.
- CO8: The importance of psychological assessment in gauging criminal behavior. The tools and techniques required for detection of deception.

BFC-131 Forensic Dermatoglyphics and Technological Methods in Forensic Science

Course outcome:

- CO1: The fundamental principles on which the science of fingerprinting is based.
- CO2: Fingerprints are the most infallible means of identification.
- CO3: The world's first fingerprint bureau was established in India.
- CO4: The method of classifying criminal record by fingerprints was worked out in India, and by Indians.
- CO5: The physical and chemical techniques of developing fingerprints on crime scene evidence.
- CO6: The significance of foot, palm, ear and lip prints.
- CO7: The importance of chromatographic and spectroscopic techniques in processing crime scene evidence.
- CO8: The utility of colorimetry, electrophoresis and neutron activation analysis in identifying chemical and biological materials.
- CO9: The significance of microscopy in visualizing trace evidence and comparing it with control samples.
- CO10: The usefulness of photography and videography for recording the crime scenes.

BFC-141 Forensic Chemistry and Forensic Biology

Course outcome:

- CO1: The methods of analyzing trace amounts of petroleum products in crime scene evidence.
- CO2: The methods of analyzing contaminants in petroleum products.
- CO3: The method of searching, collecting, preserving and analyzing arson evidence.
- CO4: The classification of explosives, including the synthesis and characterization of representative analogs.
- CO5: The significance of bomb scene management.
- CO6: The techniques of locating hidden explosives.
- CO7: The classification and characteristics of the narcotics, drugs and psychotropic substances.
- CO8: The significance of biological and serological evidence.
- CO9: The forensic importance of hair evidence.
- CO10: The importance of biological fluids – blood, urine, semen, saliva, sweat and milk – in crime investigations.
- CO11: How wildlife forensics aid in conserving natural resources.
- CO12: How forensic entomology assists in death investigations.

BFC-151 Forensic Ballistics and Forensic Toxicology

Course outcome:

- CO1: The classification of firearms and their firing mechanisms.
- CO2: The methods of identifying firearms.
- CO3: The characteristics of ammunition.
- CO4: The importance of firearm evidence.
- CO5: The nature of firearm injuries.
- CO6: The methods for characterization of gunshot residue.
- CO7: The significance of toxicological studies in forensic science.
- CO8: The classification of poisons and their modes of actions.
- CO9: The absorption of poisons in body fluids.
- CO10: The forensic identification of illicit liquors.
- CO11: The classification and characteristics of the narcotics, drugs and psychotropic substances.
- CO12: The menace of designer drugs.
- CO13: The methods of identifying and purifying narcotics, drugs and psychotropic substances.

BFC-161 Forensic Anthropology and Forensic Medicine

Course outcome:

- CO1: Importance of forensic anthropology in identification of persons.
- CO2: Different techniques of facial reconstruction and their forensic importance.
- CO3: Significance of somatoscopy and somatometry.
- CO4: The duties of the first responding officer who receives a call on homicide or suicide case.
- CO5: The steps involved in processing the death scene.
- CO6: The importance of ascertaining whether the crime was staged to appear as suicide or accident.
- CO7: The importance of bloodstain patterns in reconstructing the crime scene.
 - a. The importance of autopsy.
 - b. The importance of forensic odontology

Masters of Science (Biotechnology)

Programme Outcome (POs)

- PO1. To understand the basic unit of the organism. To differentiate the organisms by its cell structure. To know Components of the Cell and their division.
- PO2. To explain the arrangement of Genes and their interaction.
- PO3. To describe the influence of environment on gene expression. To understand extra nuclear inheritance, linkage & crossing over.
- PO4. The diversity of metabolic processes occurring in biological system. The effect of the structural and functional role of the enzymes governing the metabolic processes.
- PO5. Importance of the metabolic pathways in maintaining homeostasis in biological system. The clinical implications of the metabolic pathway.

Programme Specific Outcome

- PSO1. To explain genome organization in higher organisms.
- PSO2. To describe kinetic classes of DNA and Gene families.
- PSO3. To understand the steps involved in recombinant DNA technology.
- PSO4. To explain the construction of DNA & c DNA library and their applications.
- PSO5. To get insight in Primary and Secondary organs of Immune system.

Course Outcomes:

1ST Sem.

MBT 511 CELL BIOLOGY

- CO1. To learn about genetic engineering, principles involved in manipulating genes and DNA.
- CO2. To know about cloning strategies and expression systems.
- CO3. To acquire basic understanding of techniques in genetic engineering.

MBT 512 BIOMOLECULES AND METABOLISM

- CO1. An understanding of the intersection of life and information sciences, the core of shared concepts, language and skills.
- CO2. The ability to speak the language of structure-function relationships, information theory, gene expression, and database queries .
- CO3. After successful completion of the course student will be able to understand Principles and applications of microscopy Principles and applications of centrifugation

MBT 513 MICROBIAL PHYSIOLOGY AND GENETICS

- CO1. Applications of different spectroscopic techniques in biology Applications of chromatographic techniques in biology Principle and application of electrophoresis and blotting
- CO2. Nature and types of radiations and their application in structural and functional analysis of biological samples.
- CO3. To learn about genetic engineering, principles involved in manipulating genes and DNA.

MBT 514 IMMUNOLOGY

- CO1. After successful completion of the course student will be able to understand Isolation and quantification of nucleic acids (DNA and RNA).
- CO2. Amplification of DNA using polymerase chain reaction.
- CO3. Principles of nucleic acid hybridization and blotting techniques Methods of DNA sequencing Procedure of gene manipulation and cloning Use of Restriction endonucleases in gene cloning.

2nd Sem.**MBT -521 BIOINFORMATICS**

- CO1. Knowledge and awareness of the basic principles and concepts of biology, computer science and mathematics existing software effectively to extract information from large databases.
- CO2. To use this information in computer modeling problem-solving skills, including the ability to develop new algorithms and analysis methods.
- CO3. To know about cloning strategies and expression systems. To acquire basic understanding of techniques in genetic engineering.

MBT -522 TOOLS AND TECHNIQUES IN BIOTECHNOLOGY

- CO1. After successful completion of the course student will be able to understand Scope of animal biotechnology Introduction to the balanced salt solutions and growth medium for animal cell and tissue culture
- CO2. Basic techniques of mammalian cell cultures in vitro Biology and characterization of the cultured cells Measuring parameters of growth Applications of animal cell culture.
- CO3. After successful completion of the course student will be able to understand Concept of cellular to tipotency

MBT -523 GENETIC ENGINEERING

- CO1. Introduction to the techniques of plant tissue culture Plant genetic engineering Agrobacterium mediated gene transfer in plants
- CO2. Molecular mechanism of photosynthesis and nitrogen fixation Greenhouse technology.
- CO3. Understand terms in cancer biology, Causes of cancer. The technologies used in cancer research and diagnosis

MBT 524 BIOPROCESS ENGINEERING AND TECHNOLOGY

- CO1. Understand the underlined concepts of Biostatics, Bioinformatics, Ethics and patenting and its application in the field of Biotechnology.
- CO2. Understand the concepts of Prokaryotic and Eukaryotic Transformation and PCR.
- CO3. Describe the expression of heterologous genes and the vectors involved in it.

3rd Sem.**MBT 531 ANIMAL BIOTECHNOLOGY**

- CO1. Appreciate technology involved in industrial products of Protein engineering. Explain and illustrate techniques like Phage display and gene therapy.
- CO2. Develop the critical thinking ability and communication skills. Understand and apply the scientific method.
- CO3. Develop the aptitude to work on a scientific problem and look for alternative solution.

MBT 532 PLANT BIOTECHNOLOGY

- CO1. Enzyme engineering, design and construction of novel enzymes.
- CO2. Application of enzymes in medicine (therapeutic enzymes, enzymes as analytical reagents), drug synthesis and biosensors.

MBT 533 HUMAN GENETICS AND HUMAN GENOME

- CO1. The different types of stem cells, how they are derived and the extent of their plasticity.
- CO2. To describe antibody-antigen interaction, autoimmune diseases

MBT 534 SOCIAL, ETHICAL, LEGAL AND MANAGEMENT ISSUES

- CO1. Write their finding in a form of a thesis and defend it by presenting it in front of their teachers and examiners.
- CO2. Experience and embrace the habit of ethical practice in performing experiments and communicating them.

4th Sem.

MBT 541 ENZYME AND ENZYME TECHNOLOGY

- CO1. After successful completion of the course student will be able to understand Introduction, nomenclature and classification of enzymes.
- CO2. Enzyme kinetics and mechanism of action of enzymes
- CO3. Enzyme technology: immobilization of enzymes, commercial production of enzymes.

MBT 542 STEM CELL BIOLOGY & SOMATIC AND GERM CELL ENGINEERING

- CO1. How tumor stem cells give rise to metastases.
- CO2. Treatment-resistant remnant cells that cause relapse.
- CO3. How this impacts on the development of future cancer treatment strategies.

MBT 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research.

Masters of Science (Botany)

Programme Outcome (POs)

- PO1. Students know about different types of lower & higher plants their evolution in from algae to angiosperm & also their economic and ecological importance.
- PO2. Cell biology gives knowledge about cell organelles & their functions
- PO3. Molecular biology gives knowledge about chemical properties of nucleic acid and their role in living systems.
- PO4. Genetics provides knowledge about laws of inheritance, various genetic interactions, chromosomal aberrations & multiple alleles.
- PO5. Structural changes in chromosomes.
- PO6. Student can describe morphological & reproductive characters of plant and also identified different plant families and classification.
- PO7. They know economic importance of various plant products & artificial methods of plant propagation
- PO8. Use modern Botanical techniques and decent equipment's.
- PO9. To inculcate the scientific temperament in the students and outside the scientific community.

Programme Specific Outcome

- PSO1. Students acquired knowledge through practical work in fields as well as in laboratory.
- PSO2. Students are exposed to various industrial processes by industrial training.
- PSO3. Project helps for creating research attitude among the post graduate students
- PSO4. To create awareness about cultivation, conservation and sustainable utilization of biodiversity.
- PSO5. To know advanced techniques in plant sciences like tissue culture, Phytoremediation, plant disease management, formulation of new herbal drugs etc.
- PSO6. Students able to start nursery, mushroom cultivation, biofertilizer production, fruit preservation and horticultural practices.

Course Outcomes:

1ST Sem.

MSB 511 BACTERIOLOGY, VIROLOGY & GENERAL MICROBIOLOGY

- CO1. Students will be able to understand the structure, type and identification of Bacteria and cyanobacteria.
- CO2. Students will gain understanding of Thallus structure, reproduction and economic importance algae.
- CO3. Students will gain understanding of the classification, structure of mycelium reproduction of fungal species.

MSB 512 BIOLOGY AND DIVERSITY OF FUNGI AND PLANT PATHOLOGY

- CO1. They will know about the hazardous and useful fungi. Student will also know and learn classification and evolutionary trends in fungi.
- CO2. The plant diseases, causal organism, host and their relationship and control measures for plant diseases.

- CO3. Understanding of fungicide and use of chemical physical and biological controlling of diseases mentioned in the unit.

MSB 513 BIOLOGY AND DIVERSITY OF ALGAE, BRYOPHYTES AND LICHENS

- CO1. The characters, distribution, classification and regeneration in Bryophytes.
CO2. The characters of different orders of Bryophytes.
CO3. How the stele evolution occurs in Pteridophytes and also familiar with the work done by Indian pteridologist.
CO4. The classification of Pteridophytic classes and the morphological and anatomical characters of genus included in the different Pteridophytic order.

MSB 514 BIOLOGY AND DIVERSITY OF PTERIDOPHYTES AND GYMNOSPERMS

- CO1. Students can critically differentiate fossil and living fossil.
CO2. Students will also understand the evolutionary tendencies and comparative morphology of Cycadales, Cycadeodales and Pteridospermales.
CO3. Students can compare the characters of different orders & relationship of each order from Cordaitales to Gnetales.
CO4. Student can critically differentiate the characters of three orders of Gymnosperm i.e., Ginkogales, Coniferales, and Taxales.

2nd Sem.

MSB 521 ANGIOSPERM ANATOMY, EMBRYOLOGY AND PALYNOLOGY

- CO1. Student will understand floral structure of Angiospermic plants and how stamens and carpels are evolved.
CO2. They will also understand adaptive feature of pollinators, will get to know about scope, aim, principles of taxonomy. They will get knowledge about concepts of taxa, genus etc.
CO3. Students will get knowledge about various taxonomic evidences. They will also understand how to prepare herbarium sheets and how to read floras.

MSB 522 ANGIOSPERM MORPHOLOGY AND TAXONOMY

- CO1. Students will develop skill and will be able to prepare staining of salivary gland chromosomes of Chironomous larva. Students will be able to understand the isolation of chloroplast.
CO2. Students will gain knowledge about floral symmetry and anatomical features of various taxa.
CO3. Student can also draw good diagrams of pollen types and stamens and carpels.

MSB 523 WATER RELATIONS, GROWTH AND DEVELOPMENT

- CO1. Students will understand about biosystematics. They will also understand adaptive features of ICBN.
CO2. Students gain the skill of identifying the fossil specimen. Understands the action of low treatment of colchicine and para-dichlorobenzene to plant tissue creating polyploidy in the organism.
CO3. Students will understand the importance of photosynthesis in plants. They will also understand photosynthesis is one of the most important processes that allow plants to Live.

MSB 524 CYTOLOGY AND MOLECULAR BIOLOGY OF PLANTS

- CO1. Understanding of the structure of chromosome and how the packaging of DNA occurs. Student can differentiate Euchromatin and heterochromatin region of chromosome on the basis of staining properties.
- CO2. Student can draw a good karyotype and Idiograms of Karyotype, and also how the evolution of Karyotype takes place.
- CO3. Students will have developed knowledge about nucleus and its ultrastructure. They will also identify various forms of DNA. Student will understand the importance of stresses in plants and how it responds.

3rd Sem.**MSB 531 BIOCHEMISTRY AND METABOLISM IN PLANTS**

- CO1. Overview of major biomolecules –carbohydrates, lipids, proteins, aminoacids, nucleic acids, classification, structure, function.
- CO2. Significance and biosynthesis and the degradation pathways of the above mentioned biomolecules.
- CO3. Student will understand importance of metabolism to maintain living state of cells.

MSB 532 PLANT BREEDING

- CO1. Plant reproductive parts development of male, female gametophytes and fruits.
- CO2. Vascular tissues and its constituents by sections and maceration, wood anatomy, TS, TLS□ and RLS Mechanical tissues (Collenchyma, Sclerenchyma, Stone cells and Xylem) , Secretary tissues (Mucilage Canals, Resin canals, Nectaries, and oil glands), laticifers (Latex cells and Vessels).
- CO3. Normal and abnormal secondary growth etc.

MSB 533 PLANT BIOTECHNOLOGY: IN VITRO CULTURE, GENETIC ENGINEERING AND IPR ISSUE

- CO1. They also understand role of nitrogen cycle in environment.
- CO2. Students will understand how enzymes serve important function in body, in digestion and metabolism. They have developed knowledge about pathways of water through xylem and phloem.
- CO3. Know about plants anatomical structure, their developmental patterns

MSB 534 HERBAL TECHNOLOGY

- CO1. Explant Culture.- Anther culture Pollen culture, Micropropagation. Embryo rescue technique. Somaclonal variation. In vitro mutation. Isolation of plant protoplasts and viability, testing.
- CO2. Ability to apply the tools and techniques of microbiology in conducting independent research.

4th Sem.**MSB 541 CROP PATHOLOGY**

- CO1. Students will have knowledge about creative genetically modified bacteria. They will get knowledge that advance proteomic technologies can help us to develop better drugs.
- CO2. Students will know how they can grow disease free plant by tissue culture technique. They will develop understanding about how gene technology has helped in improving various qualities in Crops.

- CO3. Students will know about the use of computational approach to analyze, manage & store biological data. They are able to know, the use of information technology in biotechnology for data storage, Analyzing the DNA sequences.

MSB 542 ECOLOGY-I CLIMATOLOGY, SOIL SCIENCE AND AUTECOLOGY

- CO1. On completion of this course the students are able to analyze various types of ecosystems, correlate different ecosystems. To analyze the threat and suggest conservative measures.
- CO2. The students are also trained in the environmental impact analysis
- CO3. Students are able to analyze, monitor various physical, chemical and biological properties of soil water and air.

MSB 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. It is one of the small scale industry which can provide the student employment opportunity

Masters of Science (Chemistry)

Programme Outcome (POs)

- PO1:** Students gained functional knowledge of the fundamental theoretical concepts and experimental methods of Chemistry.
- PO2:** Students acquired knowledge to analyze on unknown organic compound by using spectroscopy and synthesize the compound by synthetic approach.
- PO3:** They learned to handle the instruments like TLC, HPPL, UV Spectro photo meter etc.
- PO4:** They explored various job opportunities in the Govt. sectors such as IOCL, NPTC, ONGC, DRDO, ISRO, Forensic laboratories, AP GENCO etc. and the knowledge of research area within Chemistry.

Programme Specific Outcome

- PSO1:** Understand the Concept of Aromaticity to Organic Compounds and Natural Products
- PSO2:** Explain the properties of metal Complexes and Transition Elements.
- PSO3:** Analyze the indicators and Sampling techniques.
- PSO4:** Use spectrophotometer to analyze known and unknown organic and natural compounds. Through UV, IR NMR Mass data
- PSO5:** Applies the uses of drugs and Antibiotics in Daily life.

Course Outcomes:

1st Sem.

MCH-511 INORGANIC CHEMISTRY

- CO-1** Determine and Learn about Dipole moment and bond order of the inorganic molecule.
- CO-2.** Learn about geometry and shape of the molecule.
- CO-3.** Known the preparation and properties of transition metal carbonyls
- CO-4.** To understand the 18 electron rule and its application.
- CO-5.** Find out the point group of inorganic molecules.
- CO-6.** Learn molecular orbital and its orientation.
- CO-7.** Learn concept of symmetry elements in molecules.

MCH-512 ORGANIC CHEMISTRY-I

- CO-1.** Learn SN1, SN2 and SNi Mechanism and stereochemistry.
- CO-2.** Learn classical and non-classical carbocation, NGP by pi and sigma bonds.
- CO-3.** Solve the elimination problems.
- CO-4.** Distinguish between type of addition, elimination and substitution reaction.
- CO-5.** Learn E and Z nomenclature in C, N, S, P containing compound, Stereo chemical principal, enantiomeric relationship R and S.

MCH-513 PHYSICAL CHEMISTRY-I

- CO-1.** Realize the terms ionic strength, activity coefficient, DHO equation.

CO-2. Know the Eigen function, Eigen value, operator and postulates of quantum mechanics.

CO-3. Learn two and three dimensional box, mechanics of particle.

Group Theory & Vibrational Spectroscopy MCH-514

CO-1. Study the various name reaction with examples.

CO-2. Learn the mechanism of rearrangement reaction, use synthetic reagent of oxidation and reduction for solving the problems.

CO-3. Understand the factors affecting UV-absorption spectra, Interpret IR- spectra on basic values of IR-frequencies.

CO-4. Discuss the problem of UV, IR and NMR.

MATHEMATICS FOR CHEMISTS MCH-515 (a)

CO-1. Study ^1H NMR Spectroscopy: Chemical Shift, deshielding, correlation for protons bonded to carbon and other nuclei.

CO-2. Study of ^{13}C NMR spectroscopy: FT- NMR, type of ^{13}C NMR spectra, proton decoupled, off resonance, APT, INEPT, DEPT, Chemical shift, nuclear and hetero nuclear coupling constant

CO-3. 2D NMR techniques: COSY, homo and hetero nuclear 2D resorts spectroscopy, NOESY and the applications

CO-4. Study of mass spectrometry: Instrumentation, various methods of ionization, SIMS, FAB, MALDI. Different detectors rules of fragmentations of different functional groups.

MCH-515 (b) BIOLOGY FOR CHEMISTS

CO1. Students will have a firm foundation in the fundamentals and application of current chemical and scientific theories including those in Analytical, Inorganic, Organic and Physical Chemistries.

CO2: Understand the common themes running through ionic, covalent and metallic descriptions of chemical bonding, including principles of main group elements. Enhance the knowledge on metal clusters and nuclear chemistry.

2nd Sem.

MCH-521 INORGANIC CHEMISTRY -II

CO-1. Understand the mechanism in transition metal complexes, Born Haber cycle to calculate lattices energy.

CO-2. Learn the use of catalyst, radius ratio rule of coordination number 3, 4.

CO-3. Study the structure of atom, Hunds rule, term symbol, calculation of microstate and selection rule.

CO-4. Understand the metal complexes in biological system.

MCH-522 ORGANIC CHEMISTRY -II

CO1. Explain the fundamentals of atomic structures with respect to quantum mechanical approach in detail.

CO2. Understanding wave mechanics in three dimensions and able to discuss about the advanced concepts of chemical kinetics.

MCH-523 PHYSICAL CHEMISTRY -II

CO-1. Learn the thermodynamic description of exact, inexact differential and state function.

CO-2. Know the qualitative properties of solution, the depression in freezing point, elevation in boiling point and osmotic pressure.

CO-3. Know the statistical thermodynamics and various partition functions.

MCH-524 Magnetic Resonance and Mössbauer Spectroscopy

CO1. Illustrate the principles behind the Metal Ligand equilibria in solution with respect to the formation, their Kinetic and thermal stability, and determinations, crystal field theory of transition metal complexes in octahedral and tetrahedral geometry.

CO2. Predict and account for the most commonly encountered reaction mechanisms in organic chemistry including aromatic substitution reaction, addition reactions, elimination reactions and rearrangements as well as basics of amino acids and peptides.

CO3. Develop skills in numeracy and problem solving. The subject specific skill is the acquisition of a theoretical framework which underlies much of spectroscopy.

MCH-525 Computational Methods in Chemistry

CO1. Understand concepts of partial molar properties, concept of fugacity and their determination methods including Debye-Huckel theory to strong electrolytes and also learn the thermodynamics of electrified interface.

CO2. Recognize symmetry elements in a molecule; State the point group a molecule belongs to;

CO3. Understand the role of symmetry in electronic spectroscopy, selection rules;

MCH-526 Computational Methods in Chemistry

CO1. Have the core idea about advanced organic chemistry principles and theories to develop research oriented skills in applied organic chemistry.

CO2. Understand the concept and definitions of Aliphatic nucleophilic and electrophilic substitution reactions, fundamentals of free-radicals, pericyclic chemistry.

CO3. In depth knowledge about organic chemical reactions with a focus on principles for effective synthetic strategies.

3rd Sem.

MCH 531 PHOTOCHEMISTRY

CO1. Describe and apply stereochemical concepts such as Chirality, stereoisomerm and stereoselectivity in relation to chemical transformations.

CO2. Encompass achieved advanced knowledge about the interactions of electromagnetic radiation.

CO3. Matter and their applications in organic spectroscopy to elucidate the structure of the organic compounds.

MCH 532 ENVIRONMENTAL CHEMISTRY

CO1. Have the core idea about advanced analytical chemistry principles and theories to develop research oriented skills in applied analytical chemistry.

CO2. Understand the concept and definitions of Aliphatic nucleophilic and electrophilic substitution reactions, fundamentals of free-radicals, pericyclic chemistry.

MCH 533 BIOINORGANIC CHEMISTRY

CO1. In depth knowledge about organic chemical reactions with a focus on principles for effective synthetic strategies.

- CO2. Describe and apply stereochemical concepts such as Chirality, stereoisomerism and stereoselectivity in relation to chemical transformations.

MCH 534 BASIC MEDICINAL CHEMISTRY

- CO1. Students will appreciate the central role of chemistry in our society and use this as a basis for ethical behavior in issues facing chemists.
- CO2. An understanding of safe handling of chemicals, environmental issues and key issues facing our society in energy, health and medicine

MCH 535 PHARMACEUTICAL BIOTECHNOLOGY

- CO1. The master's specialization, Organic Chemistry, will give you in-depth knowledge about organic-chemical reactions.
- CO2. Detailed focus on principles for effective synthesis strategies, stereo selectivity, catalysis, as well as organometallic chemistry.

4th Sem

MCH 541 ANALYTICAL CHEMISTRY

- CO-1. Study the steady state approximation Michaelis-Menten mechanism, Lindemann-Hinshelwood mechanism, chain reaction, Rate determining steps and consecutive elementary reactions.
- CO-2. Learn the molecular spectroscopy, Raman, Electronic and Mossbauer and its application.

MCH 542 CHEMISTRY OF NATURAL PRODUCTS

- CO1. Understand the adsorption of gases by solid type of isotherms
- CO2. Recognized the Fricke and ceric sulphate Dosimeter.
- CO3. Learn parent-daughter relationship, application of radioactivity, NAA, IDA. Effect of radiation and units of radiation.

MCH 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. It is one of the small scale industry which can provide the student employment opportunity

Masters of Science (Computer Science)

Programme Outcome (POs)

- PO-1. Able to developed the necessary learning skills and independence for further studies
- PO-2. Can initiate and lead projects within the scientific field and be responsible for the work of individuals and groups
- PO-3 Can communicate scientific information, challenges and findings to scholars as well as to general audience
- PO-4 Are capable of presenting and describing scientific issues and research findings in a foreign language

Programme Specific Outcome

- PSO-1 Design and develop computer programs/computer-based systems in the areas related to algorithms, networking, web design, cloud computing, IoT and data analytics of varying complexity.
- PSO-2 Acquaint with the contemporary trends in industrial/research settings and thereby innovate novel solutions to existing problems.
- PSO-3 Can make decisions in an independent, professional manner and support them
- PSO-4 Can decide which analytical methods and complex theories are applicable
- PSO-5 Can communicate statistical information.

Course Outcomes:

1ST Sem.

MCS 101 OBJECT ORIENTED PROGRAMMING USING C++

- CO1. Explain Concepts and Advantages of Object Oriented Programming.
- CO2. Apply and implement the concepts of the Object-Oriented paradigms to analyze design.
- CO3. Develop the solutions of real world problems using the Principles of information Hiding Localization and Modularity.

MCS 512 ADVANCED OPERATING SYSTEM

- CO1. Design Develop and maintain the small applications system utility for societal and academic problems using reusability concepts in team spirit.
- CO2. Demonstrate the Advanced Features of C++ Specifically Stream I/O Templates and Operator overloading and overriding.
- CO3. To design and understand the following OS components: System calls, Schedulers, Memory management systems, Virtual Memory and Paging systems.

MCS 513 EMBEDDED SYSTEM

- CO1. To evaluate, and compare OS components through instrumentation for performance analysis.
- CO2. To analyze the various device and resource management techniques for timesharing and distributed systems

MCS 514 ADVANCED DBMS

- CO1. To develop and analyze simple concurrent programs using transactional memory and message passing, and to understand the trade-offs and implementation decisions.

- CO2. The candidate will get knowledge of: Query optimization, Parallel and distributed database systems and New database architectures and query operators.

2nd Sem.

MCS 521 DATA STRUCTURES, ALGORITHMS & ANALYSIS

- CO1. To study types of NoSQL databases (Document oriented, keyValue pairs, Column-oriented and Graph)
- CO2. To understand detailed architecture, define objects, load data, query data and performance tune NoSQL databases.

MCS 522 S/W ENGINEERING & TESTING

- CO1. Able to handle large volumes of structured, semi-structured, and unstructured data using database technologies.
- CO2. Student will able to: Establish a basic understanding of the process of Database Development and Administration using MySQL.

MCS 523 ADVANCED COMPUTER NETWORKS

- CO1. Student will implement the concepts of both Operating Systems & Database Administration skills. Retrieve any type of information from a data base by formulating complex queries in MySQL
- CO2. Describe the important role of Linux operating system.
- CO3. Describe the functions of each layer in OSI and TCP/IP model.

MCS 524 NUMERICAL TECHNIQUES & DISCRETE MATHEMATICS STRUCTURE

- CO1. Explain the functions of Application layer and Presentation layer paradigms and Protocols.
- CO2. Describe the Session layer design issues and Transport layer services.
- CO3. Classify the routing protocols and analyze how to assign the IP addresses for the given network.

3rd Sem.

MCS 531 Programming in Java

- CO1. Explain and apply the Object Oriented Concepts for Solving Real Problem.
- CO2. Use the Java SDK Environment to Create Debug and Run Simple Java Programs.
- CO3. Apply Java Technology to Develop the Small Applications Utilities and Web Applications.

MCS 532 Object Oriented Analysis and Design using UML

- CO1. Describe the functions of data link layer and explain the protocols.
- CO2. Explain the types of transmission media with real time applications
- CO3. Apply Event Management and Layout Managers Using AWT Swing JDBC and Servlet for Developing the Software for Various Problems.

MCS 533 Computer Graphics

- CO1. Understand and Represent Data in Different Binary Formats
- CO2. Design Simplify and evaluate Boolean Equations and Circuits
- CO3. Explain and Analyse Basic Building Blocks of Digital Electronics and Computer

MCS 534 ELECTIVE- I

1. Digital image processing

- CO1. Design and Analyse Simple Combination & Sequential Circuits
- CO2. Analyse the Basic Computer Organisation and Programming

2. Pattern recognition

- CO1. This dissertation programme provides the candidate with knowledge, general competence.
- CO2. Analytical skills on an advanced level, needed in industry, consultancy, education and research

3. Digital signal processing

- CO1. Understand the Organisation of I/O Devices and Computer Memory Mapping.
- CO2. Able to Analyse and Design Expert Systems Through Learning the Machine

4th Sem.

MCS 541 ARTIFICIAL INTELLIGENCE

- CO1. Demonstrate and Apply Artificial Intelligence Techniques Various Types of Production Systems Characteristics of Production Systems.
- CO2. Design Neural Networks Architecture and Implement Functions and Various Algorithms Involved.

MCS 542 INTERNET COMPUTING USING ASP.NET

- CO1. Fuzzy Logic Various Fuzzy Systems and Their Functions.
- CO2. Genetic Algorithms its Applications and Advances

MCS 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Electronics)

Programme Outcome (POs)

- PO1. Learner can apply the knowledge of basic circuit law and simplify the network using reduction techniques
- PO2. Analyze the circuit using Kirchhoff's law and Network simplification theorems
- PO3. Infer and evaluate transient response, Steady state response, network functions
- PO4. Evaluate two-port network parameters.
- PO5. Synthesize one port network using Foster and Cauer Forms.
- PO6. This course prepares learner for various national level competitive examination.
- PO7. Learner can apply the knowledge of basic semiconductor material physics.
- PO8. Analyze the characteristics of various electronic devices like diode, transistor etc., and able to classify and analyze the various circuit configurations of Transistor and MOSFETs. Illustrate the qualitative knowledge of Power electronic Devices

Programme Specific Outcome

- PSO1. Learner can apply the knowledge of statistical theory of communication and explain the conventional digital communication system.
- PSO2. Apply the knowledge of signals and system and evaluate the performance of digital communication system in the presence of noise.
- PSO3. Apply the knowledge of digital electronics and describe the error control codes like block code, cyclic code.
- PSO4. Describe and analyze the digital communication system with spread spectrum modulation.
- PSO5. Infer the DC and AC characteristics of operational amplifiers and its effect on output and their compensation techniques. Elucidate and design the linear and non-linear applications of an OPamp and special application ICs
- PSO6. Illustrate the function of application specific ICs such as Voltage regulators, OTA, PLL and its application.

Course Outcomes:

1ST Sem.

MEC-511 MATHEMATICAL TECHNIQUE EC.S & NETWORK ANALYSIS

- CO1. Analyze the transmission lines and their parameters using the Smith Chart.
- CO2. Apply the knowledge to understand various planar transmission lines.

MEC 512 PHYSICS OF ELECTRONIC MATERIALS

- CO1. To expected to be familiar with Electromagnetic wave propagation and wave polarization
- CO2. Classify the Guided Wave solutions -TE, TM, and TEM, analyze and design rectangular waveguides and understand the propagation of electromagnetic waves.

MEC 513 SEMICONDUCTOR DEVICE EC.S & ELECTRONIC CIRCUITS

- CO1. Select the appropriate portion of electromagnetic theory.
- CO2. Its application to antennas. Antenna arrays and mathematically analyze the types of antenna arrays.

MEC 514 CONTROL SYSTEMS

- CO1. Design as well as conduct experiments, analyze and interpret the results to provide valid conclusions for digital modulators and
- CO2. Demodulator using hardware components and communication systems.
- CO3. Learner can derive and discuss the Maxwell's equations.

2nd Sem.**MEC 521 COMMUNICATION TECHNIQUE & NETWORKING**

- CO1. Describe the techniques used for VLSI fabrication, design of CMOS logic circuits, switches and memory.
- CO2. Describe the techniques used the design of CMOS logic circuits, switches and memory in VLSI.

MEC 522 MICROWAVE ELECTRONICS

- CO1. Students will be able to find transfer functions for given system.
- CO2. Perform modeling to analyze the physics behind semiconductor optoelectronic devices

MEC 523 ELECTROMAGNETIC & ANTENNA PROPAGATION

- CO1. Analyze discrete-time signals and systems using DFT and FFT.
- CO2. Understand and develop multirate digital signal processing systems

MEC 524 ELECTRONIC INSTRUMENTATION

- CO1. Students will be able to calculate time domain.
- CO2. frequency domain parameter for given system

3rd Sem.**MEC 531 MICROSCOPE EC.SSOR & MICROCONTROLLER**

- CO1. Learner can describe the importance of wafer fabrication process and integrated circuits and apply their applications in modern technology.
- CO2. Describe the structure and operation of MOSFETs

MEC 532 DIGITAL IMAGE PROCEC.SSING

- CO1. Describe the fundamental physical processes of optoelectronic transitions and apply the concepts to different optoelectronic devices
- CO2. Define, in depth, the principles/functionality of the most important optoelectronic devices, compare and evaluate the different device design.

MEC 533 IC TECHNOLOGY & VLSI DEC. SIGN

- CO1. Generalize the design techniques and analyze the characteristics of VLSI circuits such as area, speed and power dissipation.
- CO2. Learner can analyze discrete-time systems in both time & transform domain and also through pole-zero placement.

MEC 534 INTEGRATED & DIGITAL ELECTRONIC CIRCUITS

- CO1. Design and implement digital finite impulse response (FIR) filters.
- CO2. Design and implement digital infinite impulse response (IIR) filters.

4th Sem.

MEC 541 OPTOELECTRONICS AND OPTICAL COMMUNICATION

- CO1. Develop and utilize laboratory skills on semiconductor based device fabrication and characterization, perform data analysis and evaluate the device performance.
- CO2. Students will be able to predict stability of given system using appropriate criteria

MEC 542 RADAR, TV AND SATELLITE COMMUNICATION

- CO1. Students will be able to explain principle of operation for various sensors.
- CO2. Students will be able to describe functional blocks of data acquisition system.

MEC 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Environmental Science)

Programme Outcome (POs)

- PO1. Acquired fundamental knowledge of different aspects of environment and local, regional and global environmental problems.
- PO2. Developed environmental monitoring skills, including conduct of experiments and data analysis.
- PO3. Obtained exposure to the environmental pollution control technologies.
- PO4. Acquired the knowledge and skills needed for the environmental design and management.
- PO5. Acquired skills in the preparation, planning and implementation of environmental projects.

Programme Specific Outcome

- PSO1. At the end of this course, the student will be able to: Define and describe ecosystem and their types
- PSO2. Explain the ecological processes and their interaction with the environment Explain biodiversity, its threats and conservation methods.
- PSO3. At the end of the course, the student will be able to explain the chemical nature and interaction of the air, water and soil.

Course Outcomes:

1ST Sem.

MES-511 Fundamentals of Environmental, climatic and soil science

- CO1. To understand the meteorological parameters and its effect on earth.
- CO2. To learn about climate, its circulations and classification
- CO3. To understand mechanisms involved in oceanic circulations, marine resources and its pollution

MES 512 Ecology, Biodiversity, Forestry, Wildlife, And Their Conservation

- CO1. Effluent Treatment Plants of various Industries/Companies/Factories, Municipal Councils/Corporations,
- CO2. Central Pollution Control Board, State Pollution Control Boards, National Research Institutes/Organizations/Laboratories, NEERI, EIA, GI.
- CO3. Environmental Monitoring Projects, Environmental Consultants, Different Laboratories, NGO's.

MES 513 Environmental Economics And Natural Resources

- CO1. These should be measurable, appropriate to the course level, and phrased in terms of student achievement, not instructional or content outcomes If dual listed, indicate additional learning objectives for the higher level course.
- CO2. Wide academic and applied knowledge in biology and/or chemistry with specialization in the field of Environmental Toxicology and/or Environmental Chemistry.

MES 514 Energy And Environment

- CO1. Knowledge of the physicochemical and/or biophysical processes of importance to the natural environment.
- CO2. Research experience in a specialty that requires advanced knowledge of biology and/or chemistry through a supervised master's project that extends over several semesters.
- CO3. Knowledge of relevant methods and hypothesis testing, including experimental analysis (chemical and/or biological), statistical techniques and other tools used to analyze and solve biological and/or chemical issues in research, manufacturing, management and/or teaching.

2nd Sem.

MES 521 Air, Noise pollution and control

- CO1. Knowledge of international research in her/his specialty.
- CO2. Knowledge of international research groups in the field,
- CO3. The breadth of research being done in the fields of "Environmental Toxicology" and "Environmental Chemistry".

MES 522 Water, Thermal and Radioactive pollution

- CO1. To understand on formation of earth and different earth processes .
- CO2. Forest department, Water Purification and Treatment Plants and Various Sectors related to the field of Environment.
- CO3. Constitution of India and the Environment

MES 523 Hazardous waste treatment and solid waste management and occupational health

- CO1. Knowledge of chemical properties of different groups of compounds.
- CO2. Biological effects and important cycles, as basis for a comprehensive understanding of climate/environment, pollution and toxicology.

MES 524 Environmental Impact Assessment (EIA), Environmental Management System (EMS) And Environmental Audit (EA).

- CO1. Foundational basis of Environmental Law and Governance
- CO2. Principles of Environmental Governance – Legal Instruments
- CO3. Emerging Trends in Environmental Law and Governance

3rd Sem.

MES 531 Earth Processes, Hazards and Risk Assessment

- CO1. Develop a appropriate framework for research study.
- CO2. This dissertation programme provides the candidate with knowledge,

MES 532 Biotechnology , Toxicology And Environmental Management

- CO1. Waste Management Laws: Role of Municipalities and other agencies
- CO2. Forest Management in India

MES 533 Surveying, Photointerpretation And Remote Sensing

- CO1. The students have the opportunity of job and services in the field of Teaching, Researches, and Projects.
- CO2. Apply analytical tools to determine and measure pollutants in various environmental samples

MES 534 Environmental Administration, Law And Judicial Attitude

- CO1. General competence, and analytical skills on an advanced level,
- CO2. Needed in industry, consultancy, education and research.

4th Sem.

MES 541 Global Prospects Towards Environmental Ethics And Sustainable Development

- CO1. Execute sampling, collection and preservation techniques
- CO2. Develop the ability to apply the methods while working on a research project work, Describe the appropriate statistical methods required for a particular research design
- CO3. Choose the appropriate research design and develop appropriate research hypothesis for a research project

MES 542 Statistics, Biometry And Research Methodology

- CO1. Pollution Control Laws: Administrative process
- CO2. Environment Protection Act: Administrative process through delegated legislations

MES 543 Dissertation

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Food Science)

Programme Outcome (POs)

- PO1. The student will be able to utilize the various preservative methods for food in industrial settings.
- PO2. The student will comprehend the processing techniques utilized in food industries
- PO3. To study the role and significance of microorganisms in food.
- PO4. To acquaint the students with different groups of microorganisms associated with food, their activities, destruction and detection in food
- PO5. To inculcates the scientific temperament in the students and outside the scientific community.
- PO6. To understand the principles underlying the preservation methods

Programme Specific Outcome

- PSO1. To acquaint students with the chemical constituents of food, their interactions during cooking, and evaluation of taste characteristics of food .
- PSO2. To familiarize students with the classification of foods and nutrients, and their metabolism in the human body
- PSO3. To provide an optimum environment for students to gain an understanding of the chemical bases of food component reactivity and functionality.
- PSO4. To provide an opportunity for students to test various approaches for manipulating the chemical and/or functional properties of foods.8.To know about the food microbiology and in details.

Course Outcomes:

1ST Sem.

MFS 511 PRINCIPLES OF FOOD PROCESSING & PRESERVATION

- CO1. To apply knowledge of basic sciences, engineering sciences and applied engineering to design, adapt, innovate technology to process food raw materials and develop new products.
- CO2. To train students with extensive research capabilities, enabling them to propose projects to solve problems related to the food industry.
- CO3. Pin point to the scientific, technological, economic and social development of their local, regional, and national levels.

MFS 512 FOOD MICROBIOLOGY

- CO1. Students will be able to understand the importance of cereals, legumes and oil seeds and also technology used in different milling industries.
- CO2. To know technology of milling of various cereals.
- CO3. To know impart technical knowledge on refining of oilseeds

MFS 513 FOOD CHEMISTRY

- CO1. Students will be able to understand the importance of cereals, legumes and oil seeds and also technology used in different milling industries.
- CO2. To know technology of milling of various cereals.
- CO3. To know impart technical knowledge on refining of oilseeds

MFS 514 BIOCHEMISTRY AND NUTRITION

- CO1. To understand the basic composition and structure of cereals and legumes.
- CO2. To study the application of food processing and preservation principles and technologies in the processing, preservation, extension of shelf life and value addition of fruit and vegetable products
- CO3. To impart knowledge and understanding of the basic post-harvest biological, chemical, physiological and metabolic processes and changes in fruits and vegetables

2nd Sem.**MFST-521 PRINCIPLES OF FOOD ENGINEERING**

- CO1. Student will be able to understand the importance of fermentation and different microorganisms associated with foods.
- CO2. To understand the health implications of food-package interactions.

MFS-522 CEREAL AND LEGUME TECHNOLOGY

- CO1. Student will be able to understand the need for packaging food.
- CO2. To understand the various functions of food packages as influenced by their characteristics.

MFS-523 FRUIT AND VEGETABLE TECHNOLOGY

- CO1. Study the different types of microorganisms in milk and their activities - fermented dairy products and spoilage and their applications as probiotics
- CO2. To understand the health implications of food-package interactions

MFS-524 FERMENTATION TECHNOLOGY

- CO1. Understand the significance and activities of microorganisms in various foods.
- CO2. To understand the role of intrinsic and extrinsic factors on microbial growth in foods leading to spoilage.

3rd Sem.**MFS 531 TECHNOLOGY OF MEAT, FISH AND POULTRY PRODUCTS**

- CO1. To study about the processing of meat and preservation of meat by various techniques
- CO2. To study about processing of poultry and sea foods.
- CO3. To provide insight into the functions and areas of responsibility of meat inspection.

MFS 532 TECHNOLOGY OF MILK AND MILK PRODUCTS

- CO1. Student will be able to understand the technology behind the production of various dairy products.
- CO2. To impart skills in the application of biological, chemical, biochemical, physical and engineering sciences in processing and preservation of milk and milk products
- CO3. To familiarize the students with industrial standards concerning safe food production and the existent national and international systems that ensure food quality.

MFS 533 FOOD QUALITY AND SAFETY MANAGEMENT

- CO1. Recognize important microorganisms affecting food quality and safety,
- CO2. Identify methods of microorganism control to preserve food and make food consumption safe.

- CO3. Upon completion of this course, the student will be able to understand the principles and methods of Quality Control and Assurance in foods.

MFS 534 FOOD ADDITIVES, CONTAMINANTS AND TOXICOLOGY

- CO1. To understand the principles of sensory evaluation.
CO2. To understand the principles of HACCP in different food processing. (Skills) carry out sensory evaluation of a newly developed product.
CO3. Identify hazards and critical control points of different existing production processes.

4th Sem.

MFS 541 POST-HARVEST TECHNOLOGY OF PLANTATION CROPS

- CO1. Understanding status of confectionery industry in India
CO2. To learn the technologies of confectionery products.
CO3. To know about innovations in this sector.

MFS 542 TECHNOLOGY OF OILSEEDS AND FATS

- CO1. By the end of the course, the students will have Knowledge in the areas of spices processing
CO2. Student will be able to understand the need for packaging food.
CO3. To understand the various functions of food packages as influenced by their characteristics.

MFS 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Forensic Science)

Programme Outcome (POs)

- PO-1. Students know about different types of lower & higher plants their evolution in from algae to angiosperm & also their economic and ecological importance.
- PO-2. Cell biology gives knowledge about cell organelles & their functions
- PO-3. Molecular biology gives knowledge about chemical properties of nucleic acid and their role in living systems.
- PO-4. Genetics provides knowledge about laws of inheritance, various genetic interactions, chromosomal aberrations & multiple alleles.
- PO-5. Structural changes in chromosomes.
- PO-6. Student can describe morphological & reproductive characters of plant and also identified different plant families and classification.
- PO-7. They know economic importance of various plant products & artificial methods of plant propagation

Programme Specific Outcome

- PSO-1. Students acquired knowledge through practical work in fields as well as in laboratory.
- PSO-2. Students are exposed to various industrial processes by industrial training.
- PSO-3. Project helps for creating research attitude among the post graduate students
- PSO-4. To create awareness about cultivation, conservation and sustainable utilization of biodiversity.
- PSO-5. To know advanced techniques in plant sciences like tissue culture, Phytoremediation, plant disease management, formulation of new herbal drugs etc.
- PSO-6. Students able to start nursery, mushroom cultivation, biofertilizer production, fruit preservation and horticultural practices.

Course Outcomes:

1ST Sem.

MFC 511 Forensic Science, Photography, Crime Scene Management

- CO1. Understand the basic concept, meaning, significance and development of Forensic science.
- CO2. To elucidate research methodologies.

MFC 512 Criminology, Criminal Law and Police Administration

- CO1. Describe Crime scene investigations.

- CO2. Reconstruction of scenes of crime.
- CO3. Basic principles of photography and its relevance.

MFC 513 Analytical Instruments & Techniques

- CO1. To know about the concepts of Spectroscopy, electromagnetic spectrum.
- CO2. Sources of radiations, their utility and limitations.
- CO3. This paper tends to describe the different Physical instrumentation techniques.

MFC 514 Fundamentals of Questioned Document Examination

- CO1. To understand basic principles and theory of radiochemical techniques.
- CO2. To highlight general principles of biological and biochemical techniques.

2nd Sem.

MFS 521 Forensic Biology and Forensic Medicine

- CO1. To understand the history and development of Dermatoglyphics.
- CO2. Different classifications used and Modern methodologies in fingerprinting.

MFS 522 Forensic Chemistry and Toxicology

- CO1. To learn methods of taking footprints, their collection and identification of characteristics.
- CO2. To analyse lip prints and their forensic significance.

MFS 523 Forensic Ballistics and Forensic Physics

- CO1. The paper tends to introduce students to document examination.
- CO2. Also describes classification, variation, nature and problems of document examination.

MFS 524 Information Security and Cyber Crime

- CO1. To learn writing features, their estimation, general characteristics, individual characteristics, ethnic.
- CO2. Gender variability of handwriting, various types of forgeries and their detection.

3rd Sem.

MFS 531 Forensic Anthropology

- CO1. To identify typescripts, various types of printing processes and their examination.
- CO2. To Determine identification of age of documents by examination of signatures, paper, ink etc.

MFS 532 Forensic Dermatoglyphics and other impressions

- CO1. To understand concept of crime and recent development in its control and prevention.
- CO2. To study the aim and scope of criminology.

MFS 533 Forensic Genetics & Forensic Serology

- CO1. To elucidate Criminal profiling and modus operandi, portrait parley, voice stress analysis
- CO2. To describe History and development of police administration.

MFS 534 Forensic DNA Profiling & Bioinformatics

- CO1. Police duties, responsibilities and powers.

4th Sem.

MFS 541 Forensic Drugs Analysis

- CO1. This paper furthermore addresses the use of biometric evidences.
- CO2. Use modern Botanical techniques and decent equipment's.

MFS 542 Analytical Forensic Toxicology

- CO1. Techniques used in the formation of research design on a specific problem.
- CO2. To inculcates the scientific temperament in the students and outside the scientific community.

MFS 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Genetics)

Programme Outcome (POs)

- PO1. Students will be having knowledge about the basic concepts that form the building block for understanding Genetics i.e life cycles of Model Organisms, basic genetic experiments, mutations , polyploidy,.QTL Blood groups.
- PO2. Acquaintance of students with organization of genome and specialized chromosomes, Chromosomal theory of Inheritance, linkage, crossing over and recombination, inheritance modes in nature, maternal inheritance.
- PO3. Students would be able to understand the basic principles of Genetics using simpler organisms.
- PO4. Understanding concept & principles of genetic recombination in bacteria and bacteriophages.
- PO5. The course has been designed to expose the students of Human Genetics to understand the structure and function of cell membrane and related transport mechanisms.

Programme Specific Outcome

- PSO 1. Students will be able to understand structure, function of different cell organelles, bio molecules, understanding about the processes of cell division.
- PSO 2. Understanding about history of cytogenetics and extensions of Mendelism.
- PSO 3. Introduction to the Structure of Human Chromosomes and related conditions.
- PSO 4. Knowledge about chromosomal theory of inheritance.

Course Outcomes:

1ST Sem.

MGN 511 CONCEPTS OF GENETICS

- CO1. Course has been designed to make the student of Human Genetics Familiar with the Molecular plant world.
- CO2. The structure, shape, nutrition & genetics, as they make the backbone of Genetic Engineering and Breeding.

MGN 512 CHROMOSOMES, GENES AND GENOMES

- CO1. Students will get complete understanding about activation & differentiation of immune cells.
- CO2. The students will be able to understand role of immune system in human health.
- CO3. This core course will be focused on the allelic frequencies calculation in population genetics.

MGN 513 BIOSTATISTICS AND POPULATION GENETICS

- CO1. Behavioral genetics knowledge will be provided to the students in this course.

CO2. A complete understanding on the use of bioinformatics in Human Genetics.

MGN 514 CONCEPTS IN CELL AND MOLECULAR BIOLOGY

CO1. Knowledge about the molecular mechanisms of DNA, RNA.

CO2. Understanding of DNA damage, repair, RNA transport, post and co-transcriptional modifications, post translational modifications, RNA processing etc

2nd Sem.

MGN 521 GENE EXPRESSION AND REGULATION

CO1 Understanding of the immune system, its components, properties ,antigen, antibody & various Cytokines, role of immunity in human health & disease.

CO2 Knowledge of the genetic basis of the immune response,antibody diversity & interrelation between immunity to disease

MGN 522 HUMAN GENETICS

CO1 The course will help to develop new strategies to harness the power of immunity to defeat various diseases including cancer

CO2 Insight in the genetics of transplantation and tissue rejection,.

MGN 523 MOLECULAR PLANT BREEDING

CO1. Will have an overview of the till date developments in the field of environmental microbiology with special emphasis on the role of microbes in mitigating environment pollution.

CO2. Understands the role of microbes in bioremediation of environmental pollutants like petroleum hydrocarbons, pesticides, plastic and electronic waste.

MGN 524 INTRODUCTION TO BIOINFORMATICS

CO1. Autoimmunity and immunosuppression that can be applied to the estimates of transplant compatibility and the likelihood finding suitable donors

CO2. The course also understands utility of microbes in mineral and oil recovery.

3rd Sem.

MGN 531 RECOMBINANT DNA TECHNOLOGY

CO1. Students will be having knowledge about the mechanisms of gene expression regulation in prokaryotic and eukaryotic system

CO2 The course will help the students in learning the basics of MHC molecule and their role in human diseases

MGN 532 PLANT GENETIC ENGINEERING

CO1. The students would be briefed about the pathogenic nature of the microbes which leads to many pathological complications in human.

CO2. The course offers basics of immunology.

MGN 533 PLANT-MICROBE INTERACTION

CO1 Enable to make a research carrier in identification of new therapeutic targets for immune diseases.

CO2. Also learn the theory and practical skills in microscopy handling and staining techniques.

MGN 534 MEDICAL GENOMICS

- CO1. The course is designed with the objective to provide sound knowledge to the students about the theories of evolution.
- CO2. Drosophila Genome Project and Databases

4th Sem.**MGN 541 DROSOPHILA GENETICS**

- CO1. Drosophila genetics its importance as model for human genetic disorders,
- CO2. Mutagenesis & isolation of new variants.

MGN 542 GENETICS OF BACTERIA AND THEIR VIRUSES

- CO1. Understand the basic microbial structure and functions of various physiological groups of prokaryotes and eukaryotes.
- CO2. Understand the microbial physiology and know the various Physical and Chemical growth requirements of bacteria and get equipped with various methods of bacterial growth measurement.

MGN 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Geology)

Programme Outcome (POs)

- PO1. Geodynamics course of study designed to know about various hypothesis of origin of the Earth (Solar System).
- PO2. Broad perspective of crust, mantle and core of the Earth and reorganization of the Earth's layers.
- PO3. Knowledge of volcanism, isostasy, drifting continents, mid-oceanic ridges, island arcs earthquakes and their relation with plate tectonics.
- PO4. This course on the principles of geomorphology looks at the relationship between geological processes and evolution of landforms at a variety of scales in space and time.
- PO5. It examines endogenic processes originating within the earth like diastrophism, volcanic activity exogenic processes occurring at the earth surface by various geological agents like rivers, glaciers, sea and oceans and wind how they interact to create landforms.
- PO6. Aim of the branch 'Structural Geology' is to introduce various mega structures developed in rocks whether contemporary or epigenetic.

Programme Specific Outcome

- PSO1. Introduction to equipment and field tools to collect data for laboratory analyses.
- PSO2. Engineering properties of rocks. Importance of geology in civil engineering projects.
- PSO3. Tunnel. Geological considerations for tunneling grounds. Highway. Geological considerations for construction of highways. Dam. Geological consideration for the selection of a dam site and Reservoir.
- PSO4. Dam. Problems related to failure of Dams. Grouting. Bridge. Geological considerations for the concentration of bridges. Canal. Geological considerations and lining. Landslide: causes, effects, and prevention.
- PSO5. Geo-hazards: mitigation and management

Course Outcomes:

1ST Sem.

MSG- 511 GEODYNAMICS

- CO1. Concept of rock deformation in time and space.
- CO2. To determine possible causes of their formation and forces working behind it.
- CO3. Relation of structure with tectonic movement.

MSG- 512 GEOMORPHOLOGY

- CO1. To render understanding of interdependent nature and processes operative over earth surface.
- CO2. The concept of systems in nature and the paradigm of environmental thinking.
- CO3. Various setups of environments and the alarm of global warming.

MSG- 513 STRUCTURAL GEOLOGY

- CO1. Classification, separation, morphology, uses and geological history of foraminifers. Concept of Palaeobotany and Palynology.
- CO2. Plant life through ages. Characteristic features and assemblage of the Gondwana flora.
- CO3. Concept of naturally deteriorating system as desertification concept and quality of self-balancing systems in nature and human interference.

MSG- 514 ENGINEERING GEOLOGY

- CO1. The techniques of evaluations and the concerned impact of human development on environment systems, impact on surface and subsurface water resources.
- CO2. Understanding of energy resources exploitation and interference. Natural hazardous systems.

2nd Sem.**MSG- 521 PALEOBIOLOGY**

- CO1. Mineralogy leads to gain knowledge of composition of rocks and in-turn the earth's composition. To show how elements combine to produce crystal forms, introducing the principles of crystallography and mineralogy.
- CO2. To identify minerals; various properties or criteria are applied like physical properties, chemical composition and optical properties. Identification of most common minerals and mineral groups are introduced.

MSG- 522 GEOCHEMISTRY

- CO1. Groundwater: Origin, importance, occurrences and subsurface reservoirs. Water table contour maps.
- CO2. Geological factors governing the occurrence of groundwater. Aquifers and their classification. Broad perspective of groundwater flow and well hydraulics.

MSG- 523 ENVIRONMENTAL GEOLOGY

- CO1. Physical, Chemical and Biological characters of groundwater quality; water contaminants and pollutants. Problems of salt water intrusion in coastal aquifers, remedial measures. Radio isotopes in hydrological studies. Water harvesting. Consumptive and conjunctive use of surface and groundwater.
- CO2. Watershed management. Natural and artificial recharge of ground water. Wetland management

MSG- 524 MINEROLOGY

- CO1. Origin, composition, occurrences, prospecting and preparation of coal. Origin, migration and accumulation of petroleum and natural gas. Occurrences, association, distribution, prospecting and future possibilities of atomic minerals in India.
- CO2. Origin, mode of occurrence, association, uses and Indian occurrences of the ores of important metallic minerals. Origin, mode of occurrence, association, specification

and grades for uses in industries and Indian occurrences of important non-metallic minerals.

3rd Sem.

MSG- 531 HYDROGEOLOGY

- CO1. Application of geological knowledge in civil engineering like dams, tunnel and canal.
- CO2. Hydrological properties of rocks and occurrences of groundwater.
- CO3. Basic idea and application of photogeology in mineral exploration. Introduction to surface and subsurface mining

MSG- 532 GEOPHYSICS

- CO1. Introduction to aerial photography. Types of aerial photos.
- CO2. Geometric principles of photographs.
- CO3. Preparation of Photo-geologic Maps.

MSG- 533 APPLIED GEOLOGY

- CO1. Mosaic controlling factors of aerial photograph.
- CO2. Electro-Magnetic Spectrum. Space platforms.
- CO3. Elementary idea about active and passive sensors. Multispectral Scanners. Thematic Mappers.

MSG- 534 ECONOMIC GEOLOGY

- CO1. Concept of Geographic information system. Application of photo-geology and remote sensing in geological studies.
- CO2. Processes of sedimentation; Origin of sediments.
- CO3. Primary, secondary and organic sedimentary structures and their importance.

4th Sem.

MSG- 541 PHOTOGEOLOGY AND REMOTE SENSING

- CO1. Classification, nomenclature and characters of the common sediments- rudites, arenites and argillites and carbonate. Textures of clastic and non-clastic sedimentary rocks and their significance.
- CO2. Paleocurrent analyses and granulometric analyses of sedimentary rocks.
- CO3. Lithification and diagenesis. Elements and types of sediments depositional environments:

MSG- 542 SEDIMENTOLOGY AND SEDIMENTARY

- CO1. Concepts of sedimentary basins, provenance and facies. Tectonic framework of sedimentation.
- CO2. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research.

MSG- 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research.

Masters of Science (Microbiology)

Programme Outcome (POs)

- PO1. Understand the basic microbial structure and functions of various physiological groups of prokaryotes and eukaryotes and also learn the theory and practical skills in microscopy handling and staining techniques.
- PO2. Understand the microbial physiology and know the various Physical and Chemical growth requirements of bacteria and get equipped with various methods of bacterial growth measurement.
- PO3. Will have an overview of the till date developments in the field of environmental microbiology with special emphasis on the role of microbes in mitigating environment pollution.
- PO4. Understands the role of microbes in bioremediation of environmental pollutants like petroleum hydrocarbons, pesticides, plastic and electronic waste; also understands utility of microbes in mineral and oil recovery.
- PO5. Understand the structure, properties and function of genes in living organisms at the molecular level and knowledge about DNA as a genetic material, enzymology, and replication strategies molecular mechanisms involved in transcription and translation.
- PO6. To understand the biophysical techniques and other molecular methods. PO-9.To inculcates the scientific temperament in the students and outside the scientific community.

Programme Specific Outcome

- PSO1. Molecular Biology basically deals with study of DNA and other biomolecules essential for sustenance of life and varied mechanisms involved at molecular level.
- PSO2. Describe the importance of genetic code and wobble hypothesis and discuss the molecular mechanisms underlying mutations, detection of mutations , DNA damage and repair mechanisms Explain the concept of recombination, linkage mapping and elucidate the gene transfer mechanisms in prokaryotes and eukaryotes
- PSO3. Ability the principle of management and controls on the microbial processes in industrial settings. Ability the principles of physiological understanding in improvement of industrial processes
- PSO4. Understand the architecture of viruses,their classification and the methods used in their study. Discern the replication strategies of representative viruses from the seven Baltimore classes and comprehend the intricate interaction between viruses and host cells
- PSO5. Comprehend the role of viruses in oncogenesis, and ways of preventing/ treating viral infections.
- PSO6. Know how viruses can be used as tools to study biological processes , as cloning vectors and for gene transfer.

PSO7. To know the role in surrounding but problems.

PSO8. To know about food microbiology and in details.

PSO9. Study the different types of microorganisms in milk and their activities - fermented dairy products and spoilage and their applications as probiotics

PSO10. Understand the significance and activities of microorganisms in various food and role of intrinsic and extrinsic factors on microbial growth in foods leading to spoilage, and understand the principles underlying the preservation methods

Course Outcomes:

1ST Sem.

MMB 511 GENERAL MICROBIOLOGY

- CO1. Understand the basic microbial structure and functions of various physiological groups of prokaryotes and eukaryotes
- CO2. learn the theory and practical skills in microscopy handling and staining techniques.
- CO3. Understand the microbial physiology and know the various Physical and Chemical growth requirements of bacteria and get equipped with various methods of bacterial growth measurement.
- CO4. Appreciate the different types of fermentation processes & understand the biochemistry of various fermentations and product recovery methods. Get acquainted with techniques applicable for Improvement of microorganisms based on known biochemical pathways and regulatory mechanisms and learn the methods of immobilization of enzymes and cells.

MMB 512 MICROBIAL ECOLOGY

- CO1. Will have an overview of the till date developments in the field of environmental microbiology with special emphasis on the role of microbes in mitigating environment pollution.
- CO2. Understands the role of microbes in bioremediation of environmental pollutants like petroleum hydrocarbons, pesticides, plastic and electronic waste.
- CO3. To understands utility of microbes in mineral and oil recovery.

MMB 513 MICROBIAL GENETICS

- CO1. Understand the structure, properties and function of genes in living organisms at the molecular level and knowledge about DNA as a genetic material.
- CO2. To understand the biophysical techniques and other molecular methods.
- CO3. Enzymology, and replication strategies molecular mechanisms involved in transcription and translation.

MMB 514 ADVANCE TECHNIQUES IN MICROBIOLOGY

- CO1. Molecular Biology basically deals with study of DNA and other biomolecules essential for sustenance of life and varied mechanisms involved at molecular level.
- CO2. Describe the importance of genetic code and wobble hypothesis and discuss the molecular mechanisms underlying mutations, detection of mutations.
- CO3. DNA damage and repair mechanisms Explain the concept of recombination, linkage mapping and elucidate the gene transfer mechanisms in prokaryotes and eukaryotes

2nd Sem.

MMB 521 MOLECULAR BIOLOGY

- CO1. Ability the principle of management and controls on the microbial processes in industrial settings. Ability the principles of physiological understanding in improvement of industrial processes
- CO2. Understand the architecture of viruses, their classification and the methods used in their study.
- CO3. Discern the replication strategies of representative viruses from the seven Baltimore classes and comprehend the intricate interaction between viruses and host cells

MMB 522 MICROBIAL FERMENTATION TECHNOLOGY

- CO1. Comprehend the role of viruses in oncogenesis, and ways of preventing/ treating viral infections.
- CO2. Know how viruses can be used as tools to study biological processes, as cloning vectors and for gene transfer.
- CO3. To know the role in surrounding but problems.

MMB 523 MEDICAL MICROBIOLOGY

- CO1. To know about food microbiology and in details.
- CO2. Study the different types of microorganisms in milk and their activities - fermented dairy products and spoilage and their applications as probiotics microorganisms and acquire knowledge of the design of Fermenters and process controls.
- CO3. Develop an understanding of fermentation & inoculum media, their formulation and principles & techniques of sterilization.

MMB 524 FOOD MICROBIOLOGY

- CO1. Understand the significance and activities of microorganisms in various food and role of intrinsic and extrinsic factors on microbial growth in foods leading to spoilage,
- CO2. To understand the principles underlying the preservation methods

3rd Sem.

MMB 531 INDUSTRIAL MICROBIOLOGY

- CO1. Get acquainted with techniques applicable for Improvement of microorganisms based on known biochemical pathways and regulatory mechanisms and learn the methods of immobilization of enzymes and cells.
- CO2. Get acquainted with techniques applicable for Improvement of microorganisms based on known biochemical pathways.
- CO3. Regulatory mechanisms and learn the methods of immobilization of enzymes and cells.

MMB 532 BIostatistics AND COMPUTER APPLICATIONS

- CO1. Concepts about types of data observed in biological experiments, its handling and processing.
- CO2. It develops concepts of hypothesis and formulation of experiments.
- CO3. It gives understanding of various statistical operations needed to carry out and process the biological data.
- CO4. Concepts of different microbial metabolism like lipid and nitrogen metabolism, oxidation of fatty acid, assimilation of nitrates, ammonia assimilation.

MMB 533 MICROBIAL METABOLISM

- CO1. Learn the Processes involved in production of microbial metabolites like lipases, proteases and bacteriocins.
- CO2. To learn how to enhance the efficiency of microorganisms to produce particular metabolite and produce the same at large scale.
- CO3. Understand the background of concepts and theories in plant pathology, its principles and practical applications to disease management especially cultural, chemical and biological control methods.

MMB 534 APPLIED MICROBIOLOGY

- CO1. Ability to categorize the metabolic pathways in microorganisms and understand their roles in central metabolism.
- CO2. analyze the growth kinetics employed in industrial fermentation processes and employ in strain improvement
- CO3. Gain a working knowledge in nano biotechnology techniques and acquire the ability to use them to solve problems in bioengineering, biomedicine and agricultural/environmental issues.

4th Sem.**MMB 541 ENZYMOLOGY**

- CO1. Describe the concepts of pH, buffers, Henderson-Hasselbalch equation, biological buffer systems and their importance. Understanding the laws of thermodynamics , concepts of entropy, enthalpy and free energy changes and their application to biological systems and about various biochemical reactions and metabolic pathways.
- CO2. The students will be able to make qualitative and quantitative description of the basic enzymatic phenomena and processes.
- CO3. Understand the importance of mathematical and statistical methods required for the description, interpretation of enzymatic phenomena and processes.
- CO4. Knows the basic concepts, terms and techniques used in enzymology.

MMB 542 MICROBIAL BIOTECHNOLOGY

- CO1. Competently explain various aspects of environmental microbiology –water, air and soil and Comprehend about water pollution, Water-borne diseases and their transmission, methods of determination of sanitary quality of water and sewage treatment methods employed in waste water treatment
- CO2. Develop ability to link theoretical knowledge of enzymology with its practical application in industry, health care and environmental protection.
- CO3. Appreciate how microbiology is applied in manufacture of industrial products , learn methods in discovery of new useful
- CO4. Explain the application of biotechnology in medical and its allied fields, gene therapy , genetic counseling, acquire knowledge about antisense technology, Pharmacogenetics, Toxicogenomics, Tissue engineering, Biomolecular engineering and the impact of these novel strategies on human population, Address the bioethical issues & concerned linked to medical biotechnology.

MMB 543 DISSERTATION

- CO1. Ability to apply the tools and techniques of microbiology in conducting independent research.

Masters of Science (Physics)

Programme Outcome (POs)

- PO-1. Apply the skill and knowledge in the design and development of electronic circuits to fulfill the needs of small scale electronic industry.
- PO-2. Demonstrate, solve and an understanding of major concepts in all disciplines of physics.
- PO-3. Solve the problem and also think methodically, independently and draw a logical conclusion.
- PO-4. Employ critical thinking and the scientific knowledge to design, carry out, record and analyze the results of Physics experiments.
- PO-5. Create an awareness of the impact of Physics on the society, and development outside the scientific community.
- PO-6. To inculcate the scientific temperament in the students and outside the scientific community.
- PO-7. Use modern techniques, decent equipments and Phonics software's PO-8. Become professionally trained in the area of electronics, material science, lasers and nonlinear circuits.

Programme Specific Outcome

- PSO-1. Introduce advanced techniques and ideas required in developing area of Physics.
- PSO-2. Enhance students' ability to develop mathematical models for physical systems.
- PSO-3 Gain the knowledge of Physics through theory and practical's.
- PSO-4. Understand and apply principles of physics for understanding the scientific phenomenon in classical and quantum physics.
- PSO-5. Understand and apply statistical methods for describing the quantum and classical particles phenomenon in various physical systems.
- PSO-6. Understand good laboratory practices and safety. PSO-7. Develop research oriented skills.
- PSO-8. Make aware and handle the sophisticated instruments/equipment's.

Course Outcomes:

1ST Sem.

MPY-511 MATHEMATICAL METHODS IN PHYSICS

- CO1. The students will be able to understand and apply the mathematical skills to solve quantitative problems in the study of physics.

- CO2. Will enable students to apply integral transform to solve mathematical problems of interest in physics.
- CO3. The students will be able to use Fourier transforms as an aid for analyzing experimental data.

MPY 512 CLASSICAL MECHANICS

- CO1. The students should be able to formulate and express a physical law in terms of tensors, and simplify it by use of coordinate transforms.
- CO2. The students will be able to apply the Variational principles to real physical problems.
- CO3. The students will be able to model mechanical systems, both in inertial and rotating frames, using Lagrange and Hamilton equations.

MPY 513 CLASSICAL ELECTRODYNAMICS

- CO1. The students will be able to formulate and solve problems in quantum mechanics using Dirac representation.
- CO2. The students will be able to grasp the concepts of spin and angular momentum, as well as their quantization and addition rules.
- CO3. The students will be familiar with various approximation methods applied to atomic, nuclear and solid-state physics.

MPY 514 QUANTUM MECHANICS

- CO1. To explain and solve advanced problems based on classical electrodynamics using Maxwell's equation.
- CO2. The students will be able to analyze s radiation systems in which the electric dipole, magnetic dipole or electric quadruple dominate.
- CO3. The students will have an understanding of the covariant formulation of electrodynamics and the concept of retarded time for charges undergoing acceleration.

2nd Sem.

MPY 521 ATOMIC, MOLECULAR PHYSICS AND PLASMA PHYSICS

- CO1. The students will have an understanding of quantum behavior of atoms in external electric and magnetic fields; and become familiar with the working principle of laser.
- CO2. The students will be able to formulate basic models for electrons and lattice vibrations for describing the physics of crystalline materials.
- CO3. Develop an understanding of relation between band structure and the electrical/optical properties of a material.

MPY 522 ELECTRONICS DIGITAL CIRCUITS

- CO1. To provide an understanding of static properties of nuclei, nuclear decay modes, nuclear force and nuclear models
- CO2. To provide broad understanding of basic experimental nuclear-detection techniques
- CO3. The students will be able to grasp the concepts of spin arising naturally from the Dirac equation.

MPY 523 NUCLEAR PHYSICS

- CO1. The students will be familiar with various approximation methods applied to atomic, nuclear and solid-state physics.

- CO2. To have an appreciation for the modern aspects of equilibrium and non-equilibrium statistical physics
- CO3. Explain statistical physics and thermodynamics as logical consequences of the postulates of statistical mechanics.

MPY 524 SOLID STATE PHYSICS AND NUMERICAL METHOD & COMPUTER PROGRAMMING

- CO1. Apply the principles of statistical mechanics to selected problems
- CO2. Grasp the basis of ensemble approach in statistical mechanics to a range of situations
- CO3. To learn the fundamental differences between classical and quantum statistics and learn about quantum statistical distribution laws

3rd Sem.

MPY 531 ADVANCED QUANTUM MECHANICS

- CO1. Study important examples of ideal Bose systems and Fermi systems
- CO2. To describe the features and examples of Maxwell-Boltzmann, Bose-Einstein and FermiDirac statistics.
- CO3. To know the Quantum Statistics: Bose-Einstein and Fermi-Dirac distributions; black body radiation; Debye model.

MPY 532 STATISTICAL MECHANICS

- CO1. The students will be able to formulate basic models for electrons and lattice vibrations for describing the physics of crystalline materials; and develop an understanding of relation between band structure and the electrical/optical properties of a material.
- CO2. The student after undergoing this course will be able to: Explain different types of waveguides and their respective modes of propagation.

MPY 533 LASER

- CO1. Analyze typical microwave networks using impedance, admittance, transmission and scattering matrix representations.
- CO2. Design microwave matching networks using L section, single and double stub and quarter wave transformer.
- CO3. Explain working of microwave passive circuits such as isolator, circulator, Directional couplers, attenuators etc.

MPY 534 MATERIALS SCIENCE

- CO1. Describe and explain working of microwave tubes and solid state devices.
- CO2. Perform measurements on microwave devices and networks using power meter and VNA.
- CO3. Explain the operation of RADAR systems and recite their applications.

4th Sem.

MPY 541 MICROWAVE ELECTRONICS

- CO1. The student will develop a fundamental knowledge of nanomaterials.
- CO2. The student will demonstrate a basic understanding of the length scale that defines nano for metal and semiconductor materials.
- CO3. The student will demonstrate an understanding of the properties of materials with strong dependence on size.

MPY 542 NANOTECHNOLOGY

- CO1. The student will demonstrate an understanding of approaches to nanomaterials characterization.
- CO2. The student will demonstrate an understanding of approaches to engineering nanomaterials and nanostructures.
- CO3. The student will demonstrate an understanding of the challenges on safe nanotechnology.
- CO4. The student will demonstrate the ability to develop case studies of nanomaterials with a focus on fundamentals, fabrication, characterization, and applications.

MPY 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. The student will demonstrate the ability to write final project report to reflect his/her learning on fundamentals through the course.

Masters of Science (Zoology)

Programme Outcome (POs)

- PO-1. Student can identify and classify all Animal phylum from protozoa to Mammals, also understand the evolutionary relationship and their taxonomic aspects.
- PO-2. Knows the concept, process, physiology, and molecular basis of animal development. Also knows the methods of cultivation & economic importance of various species, honeybees, lac insects, fruit fly, Sericulture, Vermiculture etc.
- PO-3. Students know about economically important Fishery, Poultry, Animal husbandry, Goat and sheep farming and also methods of preparation and application of Milk and milk products.
- PO-4. Understand the application of Bio-pesticides; know about sources, methods and production of bio-fuel.
- PO-5. In Biotechnology student gain knowledge about various techniques such as Elisa techniques, DNA sequencing, DNA finger printing techniques, Somatic cell hybridization, cloning, Human Genome project etc.

Programme Specific Outcome

- PSO-1. Students acquired knowledge through practical work in fields as well as in laboratory.
- PSO-2. Project helps for creating research attitude among the post graduate students
- PSO-3. Students learn the basic biostatistics, experimental statistics and bioinformatics.
- PSO-4. Students understand plant organism interaction, Animal tissue culture.
- PSO-5. To inculcate the scientific temperament in the students and outside the scientific community.

Course Outcomes:

1ST Sem.

MSZ 511 LIFE AND DIVERSITY OF INVERTEBRATES

- CO1. Describe general taxonomic rules on animal classification
- CO2. Classify Protista up to phylum using examples from parasitic adaptation
- CO3. Classify Phylum Porifera to Echinodermata with taxonomic keys
- CO4. The students will be well equipped to become very competent in research or teaching fields

MSZ 512 LIFE AND DIVERSITY OF CHORDATES

- CO1. Describe Phylum Nematoda and give examples of pathogenic Nematodes
- CO2. Imparts conceptual knowledge of vertebrates, their adaptations and associations in relation to their environment.

- CO3. Classify phylum Protochordates to Mammalia. Complex Vertebrate interactions
- CO4. Basic concepts of developmental biology.
- CO4. Understanding of fishes habits and habitats and their functional anatomy

MSZ 513 BIOCHEMISTRY

- CO1. The students will be well equipped to become very competent in research or teaching fields after completion of this course.
- CO2. Interactions and interdependence of physiological and biochemical processes.
- CO3. To know define basic terms in biochemistry and biochemical techniques.
- CO4. Understanding of embryogenesis - Early development and post embryonic development

MSZ 514 CELL AND MOLECULAR BIOLOGY

- CO1. With the study of this paper students gain knowledge in the areas of responses to Systematic position, general organization and affinities of Ctenophora and Nemertea.
- CO2. Rhynchoceola; Systematic position, general organization and affinities of Rotifera;
- CO3. Systematic position, general organization and affinities of Hemichordata.
- CO4. Students gain knowledge of integumentary system - basic structure of skin, dermal and epidermal pigments, fins, and scales.

2nd Sem.

MSZ 521 ANIMAL PHYSIOLOGY

- CO1. Explain the applications of the various biochemical techniques.
- CO2. Explain the importance and applications of techniques in biochemistry.
- CO3. Explain the principle and applications of various chromatographic techniques with examples.
- CO4. Students gain knowledge in the areas of responses characterization and classification of Ostracoderms, placoderms, acanthodians, holocephali, elasmobranchs.

MSZ 522 GENETICS

- CO1. Knowledge about genetics, developmental biology and organogenesis
- CO2. Application of DNA technology and molecular biology for research
- CO3. Gains knowledge about gametogenesis, cleavage mechanisms, gastrulation and role of hormones in metamorphosis and regeneration

MSZ 523 DEVELOPMENTAL BIOLOGY AND IMMUNOLOGY

- CO1. Explain the structure and functions of various biomolecules.
- CO2. Explain the importance of tools and techniques in biology.
- CO3. Explain the principle, working, materials used and applications of electrophoresis.
- CO4. Describe the concept of light, electromagnetic spectrum and its application in absorption spectroscopy

MSZ 524 EVOLUTION

- CO1. Imparts knowledge about various metabolic and physiological mechanisms of the human body.
- CO2. Understands about neurophysiology and receptors
- CO3. Gain knowledge about hormones and bioluminescence

3rd Sem.

MSZ 531 BIOPHYSICS

- CO1. Understanding of stress physiology and endocrine mechanisms will allow them to control their stress and emotions there by diverting their energy towards the positive nation building activities.
- CO2. Provides students insight into maintaining healthy relationships with their opposite gender and allows them to make right choice about their life partner thus preventing congenital/consanguial diseases.
- CO3. Imparts knowledge regarding the various theories of evolution, evolutionary process such as variation, speciation, natural selection, origin of primates and man.

MSZ 532 MEDICAL LABORATORY TECHNIQUES

- CO1. Understanding of origin and salient features of Ostracoderms to Actinopterygii, adaptive radiation of Amphibians, Reptiles, birds and Mammals.
- CO2. Gains knowledge of functional anatomy of vertebrates from fishes to mammals .
- CO3. Understanding of evolutionary significance of internal fertilization, neoteny and paedogenesis.

MSZ 533 BIOSTATISTICS AND COMPUTER APPLICATIONS

- CO1. Students will understand the basic Concept of: Atomic & molecular structure, Thermodynamic & Bioenergetic in biological systems Structural & molecular properties of different biomolecules.
- CO2. Students get acquainted with different: Physicochemical properties & biological significance of Nucleic acids, proteins, polysaccharides, carbohydrates, vitamins, Hormones Intra molecular interactions.
- CO3. Growth curves and pyramids; sigmoid curve, J curve and hyperbola; logistic equation and concepts relating to growth
- CO4. The students will be well equipped to become very competent in research or teaching fields after completion of this course.

MSZ 534 BASIC CONCEPTS OF BIOTECHNOLOGY

- CO1. Identifies the significance of amniotic egg its structure and evolutionary significance of skeletal system.
- CO2. Students will understand: The principle, methodology
- CO3. Applications of various analytical techniques like Chromatography, Electrophoresis Spectroscopy.

4th Sem.

MSZ 541 ENVIRONMENTAL SCIENCE

- CO1. The immune system & its specificity , Various physical , biological factors affecting individual's immunity, Organs & cells involved in the development of immunity Histocompatibility & transplantation immunology Immunological techniques like immunoelectrophoresis, Immunodiffusion, counter current immuno electrophoresis, Immunoassay , Blotting techniques ELISA etc.
- CO2. Students gain knowledge about various tools & techniques used in biological systems and gives them insight about their use in research.
- CO3. Biostatistics teaches them to use the best data analysis methods in their research projects

MSZ 542 AQUACULTURE AND FARM MANAGEMENT

- CO1. Imparts knowledge to the student regarding environment and conservation biology.
- CO2. Gains knowledge in the areas of responses to Laws of limiting factor, Laws of minimum, Laws of Tolerance and Tragedy of commons
- CO3. Types of ecosystem – freshwater, marine and terrestrial,
- CO4. Population characteristics and dynamics – conceptual approach

MSZ 543 DISSERTATION

- CO1. This dissertation programme provides the candidate with knowledge, general competence, and analytical skills on an advanced level, needed in industry, consultancy, education and research. It is one of the small scale industry which can provide the student employment opportunity